



DEEP LEARNING BASED WASTE CLASSIFICATION AND RECYCLING GUIDANCE SYSTEM

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This is to certify that this project report titled “**DEEP LEARNING BASED WASTE CLASSIFICATION AND RECYCLING GUIDANCE SYSTEM**” is a bonafide record of work done by **TARUN SAHUKARI (228W1A1257), JITIN VENKATA SAI KAMINENI (228W1A1223)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

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PROJECT SUMMARY

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3	Name of The Guide	Dr. Ramesh Mande
4	Research Group	Computer Vision, Deep Learning, and Retrieval-Augmented Intelligent Systems
5	Application Area	Environmental Sustainability
6	Aim of the Project	To design and implement an intelligent system for waste classification and recycling guidance using deep learning techniques.
7	Project Outcomes	• Accurate multi-class garbage detection using deep learning. • Generation of recycling recommendations for waste materials • Improved support for waste segregation and sustainable recycling

Student Signatures

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Signature of the Guide

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LIST OF SYMBOLS

Symbol	Symbol Name
x	Input Data
$H(x)$	Mapping Function
$F(x)$	Residual Function
W_i	Convolutional Weights
W_s	Projection Weights
I	Input Matrix
K	Convolutional Kernel
z_i	Class Logit
y_i	Ground-truth Label
$\hat{y}_i$	Predicted Probability
H, W	Image Height and Width
P	Patch Size
E	Embedding Matrix
Q, K, V	Attention Matrices
d_k	Key Dimensionality
B_p, B_{gt}	Bounding Boxes
TP, FP, FN	Classification Metrics
q, d	Search Vectors

LIST OF EQUATIONS

Equation	Brief Description
(1) $H(x) = F(x) + x$	Basic residual skip connection
(2) $H(x) = F(x, Wi) + Wsx$	Residual mapping with projection
(3) $S(i, j) = (I * K)(i, j) = \sum_m \sum_n I(i - m, j - n) K(m, n)$	Convolution operation
(4) $f(x) = \max(0, x)$	ReLU activation function
(5) $P(i, j) = \max_{(m,n) \in R} X(i + m, j + n)$	Max pooling operation
(6) $P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$ (6)	Softmax class probability
(7) $L = -\sum_{i=1}^C y_i \log(\hat{y}_i)$	Categorical cross-entropy loss
(8) $x \in \mathbb{R}^{\mathbb{H} \times \mathbb{W} \times \mathbb{C}}$	Input image represented as a real-valued tensor
(9) $N = \frac{HW}{p^2}$	Total transformer patch count
(10) $z_0^p = x_p E$	Patch embedding representation
(11) $z_0 = [x_{class}; x_p^1 E; x_p^2 E; \dots; x_p^N E] + E_{pos}$	Positional encoding with class
(12) $\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$	Self-attention mechanism
(13) $F_{hybrid} = [F_r; F_v]$	Hybrid feature fusion
(14) $y = \text{Softmax}(WF_{hybrid} + b)$	Final hybrid class prediction
(15) $L = -\sum_{i=1}^C y_i \log(\hat{y}_i)$	Classification loss function
(16) $I \in \mathbb{R}^{640 \times 640 \times 3}$	YOLOv8 input image representation
(17) $D = \{(b_i, c_i, s_i)\}_{i=1}^N$	Detection output set
(18) $b_i = (x_i, y_i, w_i, h_i)$	Bounding box representation
(19) $IoU = \frac{\text{Area}(B_p \cap B_{gt})}{\text{Area}(B_p \cup B_{gt})}$	Intersection over union

(20) $L_{total} = L_{box} + L_{cls} + L_{obj}$	Total YOLO detection loss
(21) $Conf = P(Object) \times IoU$	Detection confidence score
(22) $P(c_i) = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$	Class prediction probability
(23) $Precision = \frac{TP}{TP+FP}$	Precision evaluation metric
(24) $Recall = \frac{TP}{TP+FN}$	Recall evaluation metric
(25) $F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$	F1-score evaluation metric
(26) $C = \{c_1, c_2, \dots, c_m\}$	Detected waste class set
(27) $G_j = f(c_j, R_j)$	Generated recycling guidance
(28) $Cosine\ Similarity = \frac{q \cdot d}{ q d }$	Similarity-based knowledge retrieval

ABSTRACT

Efficient waste management remains a pressing challenge due to the continued reliance on manual sorting methods that are slow, labor-intensive, and prone to inconsistencies in large-scale environments. This paper presents a multi-class garbage detection and recycling guidance system that integrates a YOLOv8-based object detection model with a Retrieval-Augmented Generation (RAG) framework to enable both accurate waste identification and actionable decision support. The proposed system processes an input image to detect multiple waste categories, including plastic, paper, glass, metal, cardboard, and biodegradable materials, along with corresponding confidence scores. These detected classes are then used to query a RAG pipeline, where relevant recycling information is retrieved from a curated knowledge base and structured into clear, step-by-step guidance using a language model. This approach ensures that the generated recommendations are context-aware, reliable, and practically useful. Experimental evaluation demonstrates strong multi-object detection performance with consistent classification accuracy across different waste types. Additionally, the system provides an intuitive interface that visualizes detection results and delivers user-friendly recycling instructions. By combining real-time object detection with knowledge-driven generation, the proposed system offers a scalable and effective solution for improving waste segregation and supporting sustainable environmental practices.

Keywords: YOLOv8, Garbage Detection, Waste Classification, Retrieval Augmented Generation (RAG), Recycling Guidance, Computer Vision, Sustainable Waste Management, Object Detection, Deep Learning, Environmental Sustainability.

Chapter 1

Introduction

1.1 Origin of the Problem :-

The background of the issue, which is discussed in this project, is the increasing challenges of solid waste management in contemporary settings. As the process of urbanization, industrialization, and the consumption of packaged goods have been on the upswing, the amount of waste produced daily has grown exponentially. This means that the waste management systems are continuously pressurized to process, sort, and recycle huge amounts of mixed waste materials on time and in a precise way. The conventional waste management methods continue to rely on the manual sorting process, in which employees can differentiate and sort materials like plastic, paper, glass, metal, cardboard and biodegradable waste through the process of visual inspection. Despite the fact that this has been practiced over many years, it is very slow, labor-intensive, inconsistent and is very susceptible to human error particularly where waste is found in cluttered or mixed states.

One of the biggest problems with manual sorting is that it is not as reliable when the waste of different categories are mixed within one scene. In real-life scenarios like in the collection of garbage in a household, municipal dumping sites, recycling centers, and common waste disposal locations, garbage is seldom presented in the form of a pure or separated phenomena. Rather, there is a combination of several objects that are partially covered, deformed, or contaminated by other materials. In these circumstances, proper identification is hard hence resulting in improper segregation. This leads to pollution of the recycle streams, low recycling efficiency and heavy environmental costs. It is noted in the paper that manual sorting has been a significant yet constrained part of most real-world waste processing systems as it is not an effective way to sort large amounts of mixed waste with accuracy and consistency.

The situation is even more critical, as it is not sufficient to determine the type of waste to promote sustainable waste management. Most of the systems and practices that are currently in place only assist in identifying or categorizing waste but they do not aid users on what to do with them after identification. That is, although a system might be able to inform the user that an object is made of plastic, glass or metal, the user still

might not be aware of how to recycle the material, clean it, and separate it or how to dispose of the material. This gives a disjunction between theoretical identification and action. This limitation is well indicated in the uploaded paper since it outlines that most waste classification methods end at detection but does not go further to decision support in the form of actionable recycling advice.

The genesis of the issue is also linked to the constraints of the previous waste classification technologies. Most traditional machine learning and deep learning systems were intended to classify single objects or simplified waste data, which are not completely reflective of real-world scenarios. Nevertheless, real-life waste management demands systems able to recognize a number of objects within a single image, localize them accurately, and deliver trusted outcomes in optically challenging scenarios. This necessity introduced the use of modern methods of object detection like YOLOv8 that could detect multiple waste items at a time with bounding boxes and confidence scores. This practical need led to the development of the project to go beyond single-object multi-class garbage detection and to multi-object, multi-class detection in the real world.

1.2 Basic definitions and Background :-

The management of waste has emerged as a significant field of study in recent years owing to the growing amount of solid wastes produced by households, industries, commercial facilities, and public places. Poor segregation and improper disposal of waste cause environmental pollution, overflow of landfills and waste of resources that could be recycled. Intelligent waste detection systems are being sought out in this regard to enhance the effectiveness of segregation and facilitate recycling in a sustainable way.

Garbage Detection :-

Garbage detection is the method of automatically detecting the waste material in an image or a video using computer vision. In contrast to manual sorting, automated garbage detection systems can sort the waste objects faster and more reliably. In the project proposed, garbage detection is applied to detect various waste products in an uploaded image.

Waste Segregation :-

Waste segregation refers to sorting waste products into various categories to ensure that waste products can be treated, recycled or discarded appropriately. One of the most significant things in waste management is proper segregation since they should be treated differently, as recyclable and biodegradable materials are not the same. The system proposed encourages this process through the categorization of waste into pre-defined categories, like paper, plastic, glass, metal, cardboard and biodegradable waste.

Object Detection :-

Object detection is the task in computer vision where the computer vision system detects the existence of objects in an image and even locates the objects. In comparison with image classification, which only predicts a single label on an entire image, object detection is able to detect multiple objects and label them separately. This renders object detection to be more appropriate in the waste management application where multiple garbage objects can be found in one image.

Bounding Box :-

A bounding box is a rectangular box which is drawn around a detected object in an image. It indicates the precise location of the object detected, and assists in differentiating between objects in multi-object scenes. Bounding boxes are employed in the proposed project to emphasize the instances of garbage detected and display the findings to the user.

Class Label :-

Class label is a category of a detected object as given by the model. In this work, every identified item of garbage is labeled with one of the accepted types of waste, e.g., paper, plastic, glass, metal, cardboard, or biodegradable. Such labels are subsequently fed to the recycling guidance module.

Confidence Score :-

A confidence score is used to show the confidence of the model that an object that has been detected is part of a predicted class. Increasing values of confidence mean more confidence in the outcome of the detection. Confidence scores are shown in the

proposed system with the class labels to enable the user to know the reliability of each prediction.

YOLOv8 :-

You Only Look Once version 8 (YOLOv8) is a deep learning-based object detector model, which is used to perform real-time recognition jobs. It localizes and classifies objects within one pipeline, thereby being fast and efficient. The primary detection model of this project is YOLOv8 as it allows detecting multiple waste items with high accuracy and has a realistic inference rate.

Multi-Class Detection :-

Multi-class detection A multi-class detector can identify and classify objects that can be in more than one category. The proposed project will be trained to identify six types of garbage, i.e. paper, plastic, glass, metal, cardboard and biodegradable garbage. This ensures that the framework can be applicable to real-world waste segregation activities where various types of materials can be mixed.

Dataset Annotation :-

The process of labeling images to enable the model to learn them in the course of training is known as dataset annotation. In object detection, annotation consists of the label of the class, as well as the bounding box coordinates of each object in the image. The garbage data adopted in this project is in Yolo V8 format that allows the model to be trained on object position and classification.

Training, validation and testing :-

They are the three fundamental steps applied in organizing data set to develop a model. The model is taught using the training set, performance is monitored using the validation set and the test set is used to test the performance of the final model on unseen data. In the proposed work, the data will be divided into 70 percent training, 20 percent validation and 10 percent testing to develop the model in a systematic manner.

Deep Learning :-

Deep learning is a subdivision of artificial intelligence that implicitly learns patterns on the basis of data by means of multi-layer neural networks. It has performed well in

image analysis and classification and object detection. The YOLOv8 garbage detector model will be based on deep learning in this project.

Retrieval-Augmented Generation (RAG) :-

In the Retrieval-Augmented Generation paradigm, the retrieval of relevant information occurs in the first stage of a knowledge source followed by its utilization by a language model to produce a grounded response. In the present project, once the garbage classes are detected the labels are relayed to the RAG module to enable the recycling related knowledge to be accessed and translated into format recommendations.

Knowledge Base :-

A knowledge base is a set of filtered information that is applied by the system to give factual and category-specific advice. The knowledge base used in the proposed project is filled with recycling instructions gathered on the basis of environmental guidelines, municipal waste documents and edited recycling materials. The RAG module uses this knowledge to create significant recycling advice.

Big Language Model (LLM) :-

A Large Language Model is an AI model that can comprehend and write human-like text. The LLM is applied in this project following retrieval in such a way that the knowledge base information on recycling can be transformed into an easy-to-understand summary. This will assist the system to give not just the results of the detection, but also the comprehensible recycling instructions.

Recycling Guidance :-

Recycling guidance is the process of giving the user a step by step guidance on how a waste material identified is to be handled. It can be cleaning, sorting, separation, recycling or appropriate disposal techniques. The recycling guidance aspect of the proposed system renders the system more feasible compared to the conventional waste detection models since it assists in making decisions after recognition.

1.3 Problem Statement with Objectives and Outcomes :-

1.3.1 Problem Statement :-

The conventional waste management systems remain heavily dependent on manual sorting whereby human labor is used in sorting waste materials and separating them. Even though this technique has been employed over a long period, it is tedious, time consuming, and lacks uniformity, particularly when bulk mixed wastes need to be handled within a short period of time. In the natural systems like households, common collection sites, recycling facilities, and dumping sites, wastes are typically presented as mixed and cluttered waste as opposed to individual wastes. In this kind of a situation, manual segregation is hard and mostly yields inappropriate classification of recyclable and unrecyclable substances. This lowers the efficiency of the recycling process in general and the probability of cross-contamination of the waste streams.

Waste identification is not the only challenge. Although the user may identify a waste item correctly, he/she may not be aware of the next correct action to take, e.g. whether the item is to be cleaned, separated, reused, recycled using a particular channel, or disposed using a special method. The majority of the current computer vision systems end at the detection or classification processes and do not offer any additional guidance to assist the users in making environmentally responsible choices. Consequently, there is an apparent disconnect between identifying technical waste and performing the actual recycling. This loophole makes the traditional waste detection systems less useful in practice in the field of sustainable waste management.

The other significant problem is that most previous methods of waste classification are tailored to simplified data or objects alone. Nevertheless, real garbage can include various materials including: paper, plastic, glass, metal, cardboard, and biodegradable waste within the same frame. An efficient system thus should be able to identify multiple objects at a time, precisely localize them using bounding boxes and then label them with the appropriate class and confidence scores. This need causes multi-object detection to be more appropriate when compared to fundamental image classification in real-life waste management activities. This requirement of a powerful multi-class garbage detection under real-life conditions is the foundation of the proposed project.

The current project aims to overcome these shortcomings by suggesting an automated system that integrates a YOLOv8-based object detection model with a Retrieval-Augmented Generation system. Here, the uploaded image is processed to identify several types of waste first and the labels of the identified classes are then forwarded to a knowledge-based module that retrieves information on the recycling and produces structured step-by-step instructions. This renders the system more helpful than the traditional detection-only models since it does not only detect the waste but also aids in making informed decisions regarding recycling. By so doing, the project will enhance the efficiency of waste segregation, sustainable disposal behavior, and offer a smart waste management assistance system.

1.3.2 Objectives :-

The primary goal that this project will undergo is creating an intuitive waste management assistive system which will be able to automatically recognize various garbage items in an uploaded image and provide recycling suggestions based on those categories. The project will be developed to integrate computer vision and knowledge-based text generation in such a way that waste recognition and recycling instructions are provided in one integrated framework.

The project objectives, in particular, are:

1. To train a YOLOv8 to detect garbage in multiple classes.

The system should be able to correctly identify and position various types of waste like paper, plastic, glass, metal, cardboard, and biodegradable items without using more than one input image. This goal aims at making it possible to detect multi-object in realistic waste conditions as opposed to single-image classification.

2. To develop a dynamic waste image upload and display user interface.

It will need a web-based interface whereby users will be able to post garbage images and see the waste objects identified and bounding boxes, class labels and confidence scores. This goal will make the system created practical, understandable, and accessible to the end users.

3. To incorporate a Retrieval-Augmented Generation module to recycle information.

Once waste is detected, the system must access material-specific recycling data in a curated knowledge base and make useful recommendations. This goal goes beyond mere detection and adds some context-aware decision support to the application.

4. To create a systematic recycling report on identified waste types.

It should not just be a list of identified materials but should include practical actions like sorting, cleaning, separation, recycling and disposal recommendations. This can enhance consciousness of the user and contribute to promoting proper waste segregation practices.

5. To develop an intelligent waste management system that can be scaled and become sustainable.

The bigger picture of the project is to offer a platform on top of which intelligent waste separation systems can be built that can be later expanded to real-time monitoring, massive scale application, and environmentally conscious waste processing applications.

1.3.3 Outcomes of the Project :-

The successful achievement of the suggested goals, both technical and practical, are reflected in the results of the project. The initial great result is that a strong multi-class garbage model was developed with the use of YOLOv8. The trained model is able to detect multiple waste materials within a single image and annotate them with a bounding box, a label and a probability. This demonstrates that it is possible to perform object-level waste recognition with this system and not image-level classification.

The second important outcome is designing the effective end-to-end application interface with the assistance of which a user will be able to upload the image of garbage and visualize the results of the detection. This kind of output makes it more usable as it is not a backend model, but is an application oriented solution that can be demonstrated and displayed in practice, in an academic setting.

The other deliverable is that a recommendation module based on recycling guidelines using a RAG is effectively incorporated. Once the types of waste are identified, the system extracts the category-specific data in the knowledge base of recycling, and converts this into structured advice using a language model. This will ensure that the

final product is more informative and helpful as it will not only inform the user about what the waste is but what to do with it.

The other significant deliverable is the one that involves the development of systematic recycling report that can be utilized to encourage the appropriate segregation practices. This system does not merely provide the raw predictions back but it develops recommendations that are significant and can add value to the knowledge that a user possesses on how to handle, sort and recycle different materials. This increases the usefulness of the system in the levels of environmental sustainability.

In terms of performance, the project gave quantifiable detection performances with Precision = 0.619, Recall = 0.451, F1-score = 0.522, mAP 50 = 0.535 and mAP 50-95 = 0.372. These values indicate that the model was trained to learn the useful visual features to identify garbage and achieve a consistent overall performance in identifying more than one class. Even though the future improvement could still be made, the results obtained prove the fact that the proposed system is efficient enough to be used as a viable foundation to automated waste detection and smart recycling assistance.

Overall, the final project result is a cohesive smart system, which includes multi-object garbage recognition, visualization of the results, retrieval-based knowledge retrieval and recycling advice generation. This makes this work more holistic than the conventional waste classification framework and the work a valuable contribution to intelligent and sustainable waste management solutions.

A more knowledgeable perspective of the waste handling process by the users is also a result of the project because the system is not only going to identify the type of material, but clearly and easily tell the end user how to recycle it. This assists in closing the breach between automated waste identification and the real world environmental decision-making.

1.4 Related Applications :-

The suggested Multi-Class Garbage Detection and RAG-Based Recycling Guidance System is applicable in various practical application fields where automatic detection of waste and guidance on correct disposal are significant. A significant related use is in smart waste sorting systems, in which mixed waste may be automatically identified and

separated into things like plastic, paper, glass, metal, cardboard, and biodegradable waste. Detection of recyclable and non-recyclable materials is more efficient and less labor intensive in computer vision based systems. Because the suggested framework is capable of detecting several waste items on one image and offers ordered recycling recommendations, it can be effectively applied in intelligent segregation settings.

A related application is in the urban solid waste management. Cities and local authorities deal with substantial amounts of mixed waste on a daily basis and monitoring the content of waste by hand is usually challenging, time-consuming and inaccurate. Such a system as the one described may aid the municipal waste work by assisting in determining what sort of garbage is picked up at the households, streets, or dumping sites and by recommending the appropriate recycling or disposal methods. This will enhance the quality of segregation, minimize the contamination of recyclable material, and contribute to more environmentally friendly urban waste processing. Both the paper and presentation place the project as a viable instrument of sustainable waste management as opposed to an academic detection model.

The project also pertains to recycling support systems that are utilized in the recycling plants and waste processing centers. Most of the current systems in this field are primarily concerned with either waste item detection or classification, particularly on conveyor belts or controlled plant areas. Nonetheless, this concept is furthered with the proposed system that incorporates the knowledge of category-specific recycling along with detection.

Chapter 2

Review Of Literature

2.1 Description of Existing Systems :-

In [1], the computer vision techniques were used to present an existing system to detect contamination in highly cluttered waste scenes. The authors demonstrated that object detection with deep learning can be effectively applied even to dense and visually challenging waste settings. This system was primarily designed to detect contamination in mixed waste sites, and hence, it is practical in measuring the real waste management environments, but its scope was more on visual recognition, as opposed to recycling advice.

In [2], an automated waste classification and sorting framework based on YOLOv8 was presented. That paper has shown that YOLOv8 can be used to quickly and efficiently detect wastes in real-time, and it is an appropriate model in automated sorting. The system demonstrated that the contemporary object detection techniques were powerful in identifying waste products with high efficiency, especially in activities where time and scalability are essential.

In [3], a waste detection process based on the YOLOv8 was presented and applied to the use in a recycling plant. The model was proposed to be applied in industrial recycling scenarios, where the classification of waste types was determined with the help of the model under controlled conditions. The paper emphasized the fact that the use of deep learning in detecting wastes can enhance efficiency in recycling plants, yet the research was confined to detection and classification.

A better YOLOv8 architecture of garbage classification recognition was introduced in [4]. The principal contribution of such a system was architectural refinements so as to enhance efficiency in classification and real-time response. The authors found that the variants of the YOLOv8, optimized to enhance detection performance, could be employed to render the model more suitable in the application with specific needs, such as fast and lightweight operation.

Another type of existing system was proposed in [5], where Retrieval-Augmented Generation and large language models were utilized to produce structured operating

procedures based on textual input. Even though the proposed work was not focused on waste detection using images, it revealed that reliable and context-aware guidance can be generated using retrieval-grounded generation. This kind of system is very relevant as it demonstrates how the knowledge-based generation can be used to aid in decision-making beyond the raw prediction outputs.

In [6], a deep learning-based system of waste classification based on the TrashNet dataset was addressed. In that research, it was demonstrated that neural network models can be trained to learn discriminative visual patterns of waste images and make them in meaningful categories. It provided a significant basis of further waste recognition systems by demonstrating that the deep learning can be effectively used to automate classification tasks in the waste management.

In [7], a computer vision and deep learning based recyclable waste category approach was suggested. The experiment showed that image recognition models could help to more accurately identify recyclable materials compared to the traditional manual approaches. This paper strengthened the role of artificial intelligence in aiding waste sorting sustainability and enhancing the classification consistency.

In [8], a small object detection and classification model based on deep learning and utilized to identify garbage waste in smart cities and IoT settings was proposed. The system underlined the applicability of intelligent waste detectors in the urban and smart-city context where its application requires fast and robotized analysis of garbage. The researchers demonstrated that deep learning could be incorporated into bigger digital infrastructure to undertake real-world environmental monitoring.

In [9], an image recognition platform to identify sustainable waste was introduced. That paper suggested the importance of deep learning neural networks in enhancing the performance of waste identification tools and promoting the practices that are environmentally friendly. The system was a manifestation of the increased attention to AI in waste segregation to minimize pollution and enhance the effectiveness of recycling.

In [10], a deep convolutional neural network was used to create a solid waste image classification system. The experiment proved the existence of CNN-based architectures that can extract significant image features of waste objects and classify them. The

system added to the literature by demonstrating that deep convolutional models can be used to substitute the classical handcrafted methods in the analysis of waste images.

In [11], a hybrid deep learning model to classify trash using transfer learning was reported. The study showed that transfer learning with the incorporation of a deep architecture has the potential to enhance the performance of trash classification systems. This is especially applicable to situations where the available training data is sparse, as the benefit of using pretrained knowledge is that it can be used to amplify learning speed and classification accuracy.

In [12], a deep learning method to perform classification of recyclable products in order to manage waste sustainably was proposed. The research highlighted the fact that AI-based classification systems can be useful in determining the products that can be recycled more efficiently and can help in achieving sustainable development objectives. The study demonstrated that automated identification of recyclable materials is a necessary process to intelligent waste management solutions.

2.2 Summary of Literature Study :-

Literature analysis demonstrates that waste management research has gradually shifted to the traditional methods of image classification to more innovative deep learning and real-time object recognition systems. Previous research found out that deep learning models are efficient at learning visual patterns on garbage images and can aid in automated classification of recyclable and non-recyclable materials more consistently than the manual approach [6], [7]. This direction was reinforced later by implementing convolutional neural networks, transfer learning, and hybrid deep learning models to enhance waste classification accuracy and resistance to varying conditions of the datasets [9]–[14].

The growing trend in the literature is the rising tendency of using YOLO-based object detection models to conduct practical waste analysis. Models built with YOLOv8 proved to be able to detect and localize multiple waste objects effectively, which are more applicable to the mixed-waste settings in the real world compared to simple image classification models [2], [3]. YOLOv8 architectural refinements were also discussed to make them faster, more accurate, and deploy more efficiently, demonstrating that object detection has become an important trend in intelligent waste segregation studies

[4]. Research on cluttered waste piles and municipal waste detection also verified that these models of waste detection can be successful in the challenging setting when the waste objects are overlapping, dense or partially covered [1], [13].

The other key observation made in the literature is that the majority of the current systems focus primarily on the performance of detection or classification, whereas few of them can be further extended to useful user-assistance after waste recognition. Despite their good technicalities in waste objects recognition, these works usually end at prediction results in terms of class labels, confidence scores or bounding boxes [8], [15]. Consequently, this will result in the user being aware of what the identified waste is, yet not how to sort, clean, recycle, or properly dispose of such waste. This poses a disconnect between identification of technical waste and practical recycling support.

Another aspect that the literature emphasizes is the importance of Retrieval-Augmented Generation and large language models to generate trustworthy, context-sensitive advice based on curated knowledge sources. A generation system based on retrieval grounded showed that retrieval of information combined with generating language is able to minimize unsupported outputs and come up with structured procedural recommendations [5]. Although that piece of work was not created to process image-based garbage detection, it offers a valuable piece of insight in this project by demonstrating how the knowledge-based generation can be incorporated into the AI systems to provide more practical decision support.

On the whole, the literature analysis shows that the current literature has reached a good level of automated waste classification, object recognition, and real-time waste recognition [1]-[4], [6]-[15]. But the majority of the previous systems can only deal with a single aspect of the problem, whether through visual recognition of waste or through the creation of knowledge-based advice independently.

Chapter 3

Proposed Method

3.1 Design Methodology :-

The design approach of the intended system is based on creating a unified system that will be capable of executing both multi-class garbage detection and recycling guidance generation within the same workflow. The approach was designed to overcome practical shortcomings of the old waste management systems, where waste segregation is typically done manually and where even those waste items that are identified are not accompanied with any serious disposal advice. To address these problems, the proposed work integrates an object detection module based on YOLOv8 with a Retrieval-Augmented Generation (RAG) module, such that the system is able to first detect waste products on an uploaded image and further offer category-specific recycling suggestions, based on a curated knowledge base.

The data gathering and preparation is the starting point of the methodology. In the case of the visual detection part, a dataset of garbage classification in the YOLOv8 annotation format was acquired on Roboflow Universe. The dataset includes annotated waste pictures that are in six large categories, namely; paper, plastic, glass, metal, cardboard, and biodegradable waste. As the system is not capable of simple classification, but rather object detection, a single image can have one or more waste objects, and each object is represented by a bounding box and a class label. Parallel to the image dataset, a different knowledge base of recycling was made ready by using the public environmental guidelines, municipal waste management sources, and edited recycling documents. This two-data representation plan is the basis of the proposed design, image-based learning is used to detect and text-based knowledge to generate recommendations.

Data preprocessing and organization comes after collection of the data set in the design methodology. Images are all downsized to a standard size of 640 x 640 pixels to fit the input size of the YOLOv8 model. The entire dataset is then split into training, validation, and testing sets with the proportion of 70:20:10, so that the model will be trainable, tuned and evaluated in a systematic manner. Image augmentation (horizontal flipping, scaling, translation, and brightness variation) is used during training to

enhance model robustness and minimize overfitting. These preprocessing tasks assist the detector to acquire helpful visual patterns despite the waste objects being in varying orientations, scales and lighting intensities. This kind of preparation is necessary since waste images in the real world tend to be both highly diverse and disorganized.

After the data is ready, the methodology continues to the YOLOv8-based detection phase. The garbage image uploaded is fed through the trained YOLOv8 model in this step that detects objects in a single-stage format. The detector locates all visible waste in the image and estimates the bounding boxes, class labels, and confidence scores associated with the waste. Using YOLOv8 is significant as it provides a decent tradeoff between detection and inference speed, thus it can be used in real-time or close to real-time. The identified waste items are subsequently visually labeled on the photograph and the user can see what materials have been identified by the model. The raw input image is converted into structured detection output in this step to be utilized in the subsequent stage of the pipeline.

The procedure is then transferred to the multi-object waste categorization and label extraction. Though YOLOv8 already offers the prediction of classes as a subset of object detection, the stage of this project involves gathering predicted categories of waste based on the model output and arranging them in such a way that they can be accessed as knowledge. When there are several objects in the image, each object is extracted with its own class detected. This is a significant design decision since the system cannot be restricted to a single waste item per image but rather it is supposed to be applied to realistic mixed-waste scenes. The framework ensures that distinct material categories are processed separately in the next recommendation phase by extracting all the identified labels.

Once the waste labels are washed out, the design approach passes through the knowledge retrieval phase of the RAG pipeline. The identified class names are then sent to a retrieval system to find the pertinent information in the prepared recycling knowledge base found in relation to a particular material. RAG module as indicated in the project presentation relies on FAISS and Google Gemini API to retrieve and generate respectively. Identification of text portions related to waste handling directions, cleaning procedures, sorting techniques, recycling procedures and disposal practices is attributed to the retrieval stage. This makes sure that the system is not

derived to create advice based on the general language modeling, but rather on material specific knowledge.

Context processing and recycling guidance generation is the next step. The retrieved textual information is then structured in this stage and delivered as contextual information to the language model. The large language model does not actually create recycling instructions but merely rephrases the information retrieved to a coherent, readable, and understandable explanation. In the case of every identified type of waste, the model will produce a step-by-step recycling instructions, which can include preparation, cleaning, sorting, separation, recycling process, and the correct way of disposing of it. This step is key to the suggested methodology since it is what brings the system outside of technical acknowledgment and transforms the output of the detection into significant user support.

The last part of the methodology is concerned with the user interface and the presentation of the results to make the system practical and user-friendly. The entire pipeline is embedded into a web application based on Streamlit, in which the user uploads a garbage picture through an interactive interface. The system shows the original image with bounding boxes, waste labels that are predicted, and their confidence scores after processing. The recycling summary generated is also structured together with the visual output. This output design makes the system user friendly even to those who may not be technically inclined to deep learning or waste management systems. The interface will then serve as the layer of interaction between the underlying AI modules and the end users in a practical, comprehensible way.

At the system level, the design approach can thus be perceived as a series of interrelated modules: dataset preparation, preprocessing, YOLOv8 training, object detection, label extraction, knowledge retrieval, recycling guidance generation, and user interface presentation. The stages are all interdependent on each other and combine to form an all inclusive end to end intelligent waste management support system.

3.2 System Architecture Diagram :-

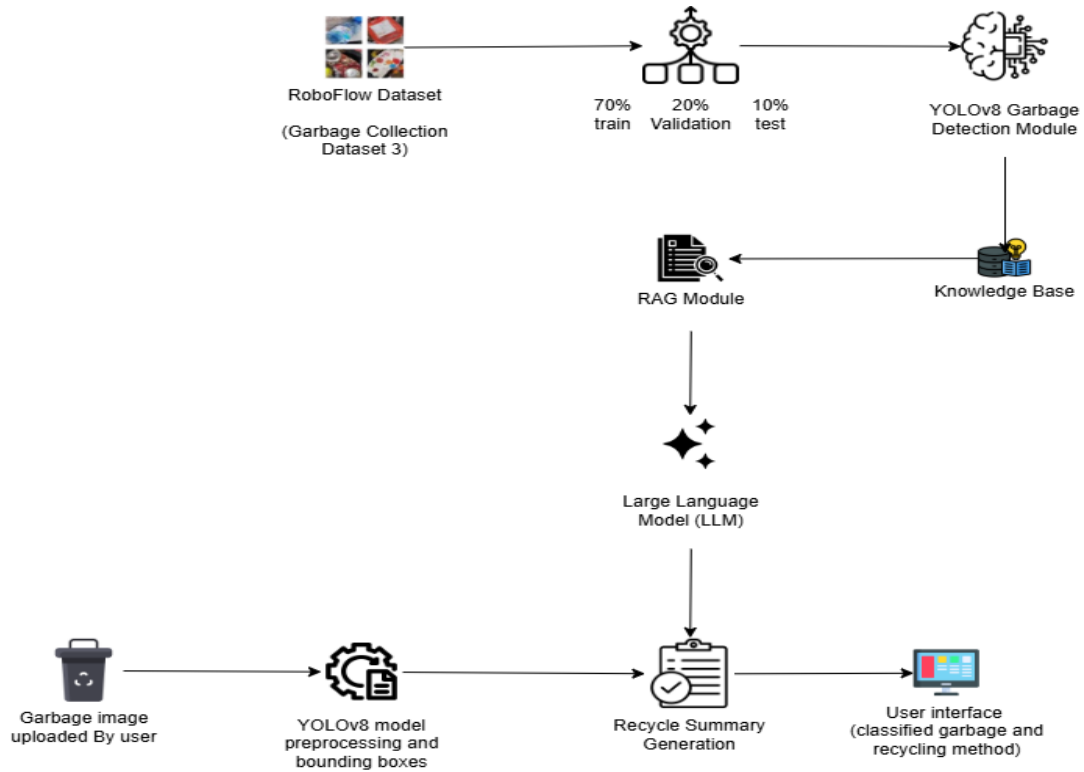


Figure 3.1: Architecture diagram of the YOLOv8 model.

The proposed framework has a system architecture in the form of an integrated pipeline, which integrates multi-class garbage detection with knowledge-based recycling guidance generation. The primary aim of such architecture is to make sure that the system does not only identify the waste objects that are found in an uploaded picture, but also offers valuable suggestions concerning the recycling and disposal of the same. The proposed architecture links computer vision, knowledge retrieval, and language generation together as an integrated system as opposed to traditional systems, which only classify or detect objects. This renders the entire system more feasible in the context of waste management applications in the real-world. The proposed architecture, with the interconnected modules as indicated in Figure 3.1, will collaborate to generate detection output and recycling assistance.

As shown in Figure 3.1, the framework may be perceived in two key phases, i.e. the model development phase and the user interaction phase. During the initial stage, waste data on Roboflow is utilized to train and test the YOLOv8 garbage detector model. The 70 percent training data, 20 percent validation data, and 10 percent testing data are used

to divide the dataset to allow the model to learn the object features, to test its performance in training and then to test its detection ability on unseen samples. This step is the learning foundation of the proposed system since the overall accuracy of identification of garbage is based on the efficiency of the trained YOLOv8 model.

Having prepared and split the databases, the YOLOv8 garbage detector is trained to detect various types of garbage like paper, plastic, glass, metal, cardboard, and biodegradable garbage. YOLOv8 is chosen as it provides a reasonable compromise between performance and detection time and can be used in multi-object recognition tasks. This module becomes the core of visual intelligence in the architecture after being trained. It is fed an image and estimates the waste classes within it and their bounding boxes and confidence scores. Therefore, the detection stage (as illustrated in Figure 3.1) carries out localization and classification in a single pipeline.

The knowledge base is another significant block presented in Figure 3.1, where the curated information related to recycling is stored concerning all waste categories supported. This warehouse is ready with ecological standards, community waste management reports, and edited recycling materials. The knowledge base serves as a factual origin of what the system gets recycling instructions. The architecture guarantees that generated output is supported by retrieved category-specific information, as opposed to the generation of responses without any grounding. Such design enhances reliability and makes the final system more beneficial in practice to waste handling purposes.

The RAG module is also linked to the knowledge base and is the module that will provide the linkage between the recommendation of recycling and waste detection. After the YOLOv8 model predicts the labels of the wastes, these labels are sent to the retrieval phase. The RAG process will scan the knowledge base to find some information pertinent to every material identified. The design allows the system to go beyond mere object detection and offer contextual decision support. The retrieved information is then sent to the next step to generate language as shown in Figure 3.1, making sure that the recommendations are specific to the identified type of waste.

The next element of the architecture is the Large Language Model (LLM). It aims at transforming the knowledge retrieved about recycling into a clear readable and organized recycling summary. The LLM does not show broken text or raw data base

entries; it formats the data into easily readable instructions. To illustrate, in case plastic is detected an output generated can be instructions on cleaning, sorting and recycling the material. Similarly, where glass is detected, the reaction can involve instructions on separation, and safe handling. Thus, as depicted in Figure 3.1, the LLM is the language-generation part, which converts technical retrieval results into useful advice.

The bottom section of Figure 3.1 shows the execution flow of the system by the user perspective in a run time. Starting with the uploading of a garbage image by the user in the application interface. The uploaded image is then transmitted to the YOLOv8 model preprocessing and bounding box phase, in which the image is processed to perform inference and all the waste objects which can be identified are detected. The materials identified, their class labels and bounding boxes are extracted and sent to the summary generation pipeline. This step is important in order to ensure that the raw image is converted to organized detection output which can be efficiently utilized by the recommendation module.

After the detection process, the system passes to the recycle summary generation stage. At this point, the results of the detection are assembled with the results of the RAG and LLM modules to come up with a meaningful recycling summary of the uploaded image. This overview encompasses not only the type of garbage that was identified but also the proposed means of recycling or disposal of each type of garbage that was identified. Therefore, it is not just that the end product is the results of technical prediction, but that it is enhanced with practical waste management information. This is among the greatest strengths of the architecture in Figure 3.1, since it directly responds to the practical requirements of the real-world to have a practical recycling advice once the waste has been identified.

The last element presented in the Figure 3.1 is the user interface where the results of the classified garbage and recycling recommendations are presented. This interface is the interface between the AI system and the end user. It shows an uploaded image with detected bounding boxes, predicted waste labels, and generated recycling instructions in a well-organized way. The user interface being executed in Streamlit makes the entire application to be interactive, easy to operate and can be demonstrated and used in practice too. Combining both detection outcomes and recycling suggestions into a

singular interface, the system can offer an entire end-to-end waste management support option.

Architecturally, the proposed structure in Figure 3.1 provides a two-level combination. The former level integrates the preparation of datasets, training, and waste detection by YOLOv8. The second level is a combination of retrieval, knowledge grounding, and language generation to support recycling. These two levels in combination form a system which is effective technically and practically useful.

3.3 Description of Algorithms :-

The general work was developed in several steps of its development, and every step added a more strict algorithmic method of garbage classification, understanding of waste, and recycling assistance. The previous phase was devoted to the classification of the garbage in terms of images with the help of a deep convolutional model. The second phase enhanced the learning of features that were based on convolutional and transformer representations. The last step took the work to the full level of intelligent framework, which could identify various waste items and produce recycling suggestions. Thus, the algorithms employed throughout the work represent a gradual shift towards classification based learning to detection based and knowledge based waste management.

3.3.1 ResNet-50 :-

ResNet-50 is a popular deep convolutional neural network that is used to perform tasks of image classification due to its capacity to learn hierarchical visual features and overcome the degradation issue in deep networks. ResNet-50 can be used in the categorization of garbage to extract discriminative local features, which include object edges, texture patterns, color distributions and surface structures. These visual characteristics can be used to distinguish types of wastes like paper, plastic, glass, metal, and cardboard.

ResNet-50 is based on the concept of residual learning. The network is trained to learn a residual mapping, i.e:

$$H(x) = F(x) + x \quad (1)$$

where x is the input to the residual block and $F(x)$ is the residual function that is being learned by the stacked convolutional layers. This skip connection allows the network to maintain gradient flow when using backpropagation, and it is used to more effectively train deeper networks.

In case the input and output dimension are unequal, the residual mapping can be expressed as:

$$H(x) = F(x, W_i) + W_s x \quad (2)$$

where W_i is the learned weights of the convolutional layers, and W_s is the projection matrix to match dimensions.

Any convolution within the network can be illustrated as:

$$S(i, j) = (I * K)(i, j) = \sum_m \sum_n I(i - m, j - n) K(m, n) \quad (3)$$

where I is the original input image or feature map, K is the convolution kernel and $S(i, j)$ is the resulting feature map. ReLU activation function is used after convolution:

$$f(x) = \max(0, x) \quad (4)$$

This brings about non-linearity and enables the network to acquire more complex visual patterns. Reduction of spatial dimensions without loss of important information is done with the help of pooling layers. Output in max pooling:

$$P(i, j) = \max_{(m, n) \in R} X(i + m, j + n) \quad (5)$$

where R represents the pooling window. The result of the final classification is obtained after the extraction of features with the help of the Softmax function:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}} \quad (6)$$

where z_i is the score of class and C is the total number of classes.

To train, the categorical cross-entropy loss will be employed:

$$L = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (7)$$

where y_i is the actual label and $\hat{y}_i$ predicts the probability. ResNet-50, therefore, offers a powerful baseline to garbage image classification through the acquisition of powerful local visual representations by the network on the dataset.

3.3.2 ViT + ResNet-50 Hybrid Model :-

ViT + ResNet-50 Hybrid Model is one of the models that has integrated the features of convolutional neural networks and transformer learning. In this design, extracting local spatial features is implemented with ResNet-50, and the global contextual relationships across the image are obtained with the help of the Vision Transformer. The idea behind merging the two models is to come up with a more featureful representation than would each of the models alone.

The ResNet-50 component is based on the residual learning, which can be once again represented as:

$$x \in \mathbb{R}^{H \times W \times C} \quad (8)$$

This branch assimilates the local characteristics like edges, shapes, fine textures and material-specific surface patterns. These characteristics are critical in classification of waste since most of the garbage types vary in the local appearance.

Vision Transformer branch starts with the division of the input image into patches. Assuming that the image is denoted by, the overall number of patches is:

$$N = \frac{HW}{p^2} \quad (9)$$

where P patch size. All patches are flattened and embedded into an embedding space:

$$z_0^p = x_p E \quad (10)$$

the flattened image patch is x_p where and E is the learnable embedding matrix.

Positional encoding is introduced to maintain position information:

$$z_0 = [x_{class}; x_p^1 E; x_p^2 E; \dots; x_p^N E] + E_{pos} \quad (11)$$

where E_{pos} is the positional embedding and x_{class} is the class token. They are then fed through layers of encoder transformers using self-attention. The self-attention operation is given to be:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (12)$$

Where (Q, K, V) , represent query and key matrices, and value matrices, and d_k represents the size of keys. This allows the model to have the long-range dependencies and general contextual structure of the picture.

The local and global features, i.e., the features of resnet-50 and viT, are then combined. Assuming that F_r denotes the ResNet-50 feature vector, and F_v denotes the ViT feature vector, then the hybrid feature representation can be determined as:

$$F_{hybrid} = [F_r; F_v] \quad (13)$$

where ; is concatenation of features. The overall prediction of the classes is:

$$y = \text{Softmax}(WF_{hybrid} + b) \quad (14)$$

Where W and b are the parameters of the classification layer that can be trained. The categorical cross-entropy loss is used to train the model:

$$L = -\sum_{i=1}^C y_i \log(\hat{y}_i) \quad (15)$$

This hybrid model enhances the performance of the classification as it involves fine-grained local feature extraction with global contextual understanding, which is more effective than a convolutional or transformer-based model alone.

3.3.3 YOLOv8 Retrieval-augmented Generation.

The last phase of the work is based on an integrated version of YOLOv8 and Retrieval-Augmented Generation (RAG). The design is based on the earlier classification-based solutions to a complete waste management support system that can recognize different garbage items, their positions with the help of bounding boxes, and generate category-related recycling suggestions.

YOLOv8 is an object detection model that is a single-stage, where localization and classification are performed in a single pipeline. Assume the input image to be:

$$I \in \mathbb{R}^{640 \times 640 \times 3} \quad (16)$$

Where I is the RGB input image of the required size. The detector comes up with a list of predictions:

$$D = \{(b_i, c_i, s_i)\}_{i=1}^N \quad (17)$$

where b_i is the predicted bounding box, c_i predicted waste class, s_i the confidence score and N the number of objects detected in the image.

Any bounding box can be represented as:

$$b_i = (x_i, y_i, w_i, h_i) \quad (18)$$

and (x_i, y_i) coordinates of the center and w_i and h_i width and height of the bounding box. Intersection over Union is used to measure the overlap of the predicted and true boxes:

$$IoU = \frac{\text{Area}(B_p \cap B_{gt})}{\text{Area}(B_p \cup B_{gt})} \quad (19)$$

where B_p and B_{gt} are the predicted bounding box and the ground-truth box.

The overall loss of detection can be defined as:

$$L_{total} = L_{box} + L_{cls} + L_{obj} \quad (20)$$

Where L_{box} is the localization loss, L_{cls} is the classification loss and L_{obj} is the objectness loss.

The model of the confidence score of detection is usually:

$$\text{Conf} = P(\text{Object}) \times \text{IoU} \quad (21)$$

The likelihood of the prediction of the class, is given as:

$$P(c_i) = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}} \quad (22)$$

Where C the number of waste classes. Evaluation of the model performance is done in the following metrics:

$$\text{Precision} = \frac{TP}{TP+FP} \quad (23)$$

$$\text{Recall} = \frac{TP}{TP+FN} \quad (24)$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (25)$$

Besides, mAP50 is calculated with IoU threshold 0.50 and mAP50-95 is an average of the precision based on the IoU threshold of 0.50-0.95. The measures are used to assess the overall performance of the garbage detector.

After the detection of the wastes is performed the predicted class labels are sent to the Retrieval-Augmented Generation module. Permission to have the waste classifications as:

$$C = \{c_1, c_2, \dots, c_m\} \quad (26)$$

in which C location are the various categories of wastes identified in the image. A retrieval system leads to a knowledge base search and retrieval of the relevant content by the retrieval system. Guidance generated can be recycled as:

$$G_j = f(c_j, R_j) \quad (27)$$

Where G_j is the generated recommendation, R_j is the retrieved knowledge and f is the language generation function.

When the embedding-based retrieval is considered, the cosine similarity is employed to measure the similarity of the query vector and a document vector:

$$\text{Cosine Similarity} = \frac{q \cdot d}{|q||d|} \quad (28)$$

This will help to reach the most suitable recycling guidelines to the material identified. Combining YOLOv8 and RAG, the framework is capable of not only performing visual recognition but also grounded recommendation, therefore, it is more realistic and practice-oriented than a conventional waste classification system.

3.4 Description of datasets, Requirements and Tools :-

The suggested large-scale project applies two sources of knowledge: a set of images to detect garbage and a text-based one to generate recycling instructions. The dataset component is not restricted to visual data alone because the system is a combination of object detection and Retrieval-Augmented Generation. Rather, a dual-data method has been used to construct the framework where the YOLOv8 detection model is trained with one dataset and the RAG-based recommendation module is trained with the other. This integration renders the system more comprehensive and viable towards smart waste management uses.

Garbage Image Dataset

The main visual data that will be employed in this project is Garbage Classification Dataset gathered on Roboflow Universe in YOLOv8 format. This data set comprises annotated garbage images that belong to six key waste categories, i.e. Paper, Plastic, Glass, Metal, Cardboard and Biodegradable waste. Given that the project is a detector of objects, each photo can have an object or objects of garbage, and each object is labeled with a class and the coordinates of the bounding box. The dataset is so that it can be used to train a multi-class object detection model as opposed to a basic image classification model.

The chosen data set is significant as it gives the indication of the practical necessity to identify various waste products that often can be found in a real-life garbage situation. Every annotation assists the YOLOv8 model to understand not only what it is an object of waste, but also where it is located in the image. This object-level learning is a necessity of the proposed system since the ultimate data should show bounding boxes, class labels, and confidence values of each identified garbage object.

Split and Organization of Data sets.

The garbage dataset is broken down into training, validation and testing sets to build the detection model in a systematic manner. The proposed project will use the split ratio of 70 percent training, 20 percent validation, and 10 percent testing. The YOLOv8 model is taught the visual appearances of various types of garbage using the training set. The validation is called in the training stage to keep track of the performance and

prevent overfitting, whereas the testing is called after training in order to consider the final performance of the detector on the unknown images. This split of data guarantees the high learning of the model and the unbiased evaluation of the performance.

Besides splitting, all pictures are also downsampled to a standard input of 640 x 640 pixels that corresponds to the requirements of the YOLOv8 model. In the training process, robustness is enhanced by applying augmentation techniques like flipping, scaling, translation and brightness change. These preprocesses are used to aid the model to generalize more in relation to various waste appearances, object orientations as well as lighting conditions.

Recycling Knowledge Base

Other than the image dataset, the project also makes use of a recycling knowledge base as the textual dataset of RAG module. This knowledge base is assembled based on guidelines of the environmental agencies, municipal waste management documentation, and edited recycling resources. It is designed to store material specific recycling data in all the waste classes supported. After the identification of the garbage labels by YOLOv8, the corresponding recycling content is extracted out of this knowledge source and, consequently, translated into structured recommendations.

The area of knowledge is critical to the entire system as the suggested project is not confined to the detection of waste only. It also seeks to offer viable recycling advice on the identified materials. The system leverages a curated textual source to make sure that the recommendations generated by the system are based on meaningful and category-specific information, as opposed to generic text generation.

Proposed System Requirement.

The software and execution environment support is needed to implement the proposed framework successfully. Because the project is a combination of computer vision, image processing, retrieval, and generation of language, the system requirements include various elements including model training, preprocessing, knowledge retrieval, and user interaction. Under the project specification, Python is used as the overall framework with the core programming language of the model training, inference and integration.

In the case of the object detection module, the project is going to need the YOLOv8 which is going to be used as the main model to identify and localize waste objects. The model is trained and tested within a Kaggle Notebook environment, which also offers the needed GPU support to run the experiment efficiently and deliver reproducible results. It specifically works well with deep learning workloads and training the detector on the prepared dataset of garbage.

The project utilizes OpenCV to process images and handle them, including image reading, image resizing, and visualising detection results. OpenCV can also be used in controlling the flow of images between the user-upload phase and the model inference phase. It thus plays the role of a vital image processing tool in the proposed architecture.

Streamlit is used to implement the web-based interaction layer of the system. This is a useful and easy to use tool allowing the user to build a front-end application that allows them to upload garbage images, see the generated results of the bounding box, and read the recycling advice. Streamlit also renders the whole system interactive and can be used to demonstrate projects and also to use it by end-users.

Tools Used in the Proposed Project

The major project employs a number of tools to help in various stages of the pipeline. The most significant one is YOLOv8 that is employed to detect garbage in multiple classes. It is the deep learning model that generates the bounding boxes, waste classes labels, and confidence of the input image. Its rapid and precise detection feature renders it apt to real-time or near real-time waste analysis.

FAISS is also employed as the retrieval engine of the RAG framework with the project. The FAISS will be used to look up the prepared recycling knowledge base and get the most pertinent information based on the identified categories of waste. This retrieval measure is to make sure that the final output generated is not ungrounded based on the available knowledge in waste handling.

The language generation component of the RAG pipeline in the project relies on the Google Gemini API, which serves as the large language model part of language generation. Having been retrieved, Gemini will reconstruct the chosen information on recycling into a clear, structured and user friendly explanation. This increases the final output to be more readable and useful to the user.

Another significant element of this framework is that the chosen datasets, requirements, and tools are not individual technical parts, but a comprehensive system that facilitates the end-to-end intelligent waste management. The image database allows the model to visually identify and localize various waste products whereas the recycling knowledgebase makes sure that the identified classes are connected with meaningful and context-specific advice on disposal. In a similar way, the software used is selected to provide support to a particular layer of the framework: Python is the programming language, YOLOv8 does the object detection, OpenCV does image processing and visualization, FAISS provides effective access to the textual knowledge base, Google Gemini API provides a structured recycling proposal, and Streamlit links all these parts together through a convenient user interface. The Kaggle Notebook environment also enhances the implementation as it supplies the computational resources necessary to train and experiment. Thus, a combination of these datasets, system requirements and tools renders the proposed framework not only technically feasible, but also scalable, interactive and applicable to the practical implementation in smart and sustainable waste management applications.

Chapter 4

Results and Observations

4.1 Stepwise description of Results :-

The part will present the results in a chronological sequence where the first one will be the classification system using ResNet-50, second will be the classification models which are hybrid and the last will be the YOLOv8 and Retrieval-Augmented Generation system. The results are arranged in a way that it is easy to understand how the image-level waste classification development to hybrid feature learning and to waste detection with contextual recycling guidance can be developed. This organization also simplifies the explanation of how the performance enhanced at every stage and how the ultimate structure became more feasible on the real-life applications of waste management.

4.1.1 ResNet-50 with Rule-Based System Results :-

The classification of the waste according to the ResNet-50 was designed in a manner that it is capable of identifying the type of waste that the input image is and then label the identified material with a recycling process through a rule-based system. This system was an initial step to automated waste analysis, replacing manual classification with a deep learning-based one. The achieved results show that the final accuracy of the model was 90.83, which demonstrates that ResNet-50 was able to extract valuable visual features to learn on waste data and make valid predictions based on classes. The accuracy of deep residual learning on the image-to-waste classification task is confirmed by the performance level of the current system and could be used as a strong basis in the future to enhance the system.

Figure 4.1 indicates that the training and validation accuracy increased with the number of epochs. The increasing trend of both curves shows the model was able to learn meaningful representations on the data and slowly built its classification ability as the training process progressed. The similarity between the two curves also indicates that the model had a reasonable generalization and did not drastically deviate during learning.

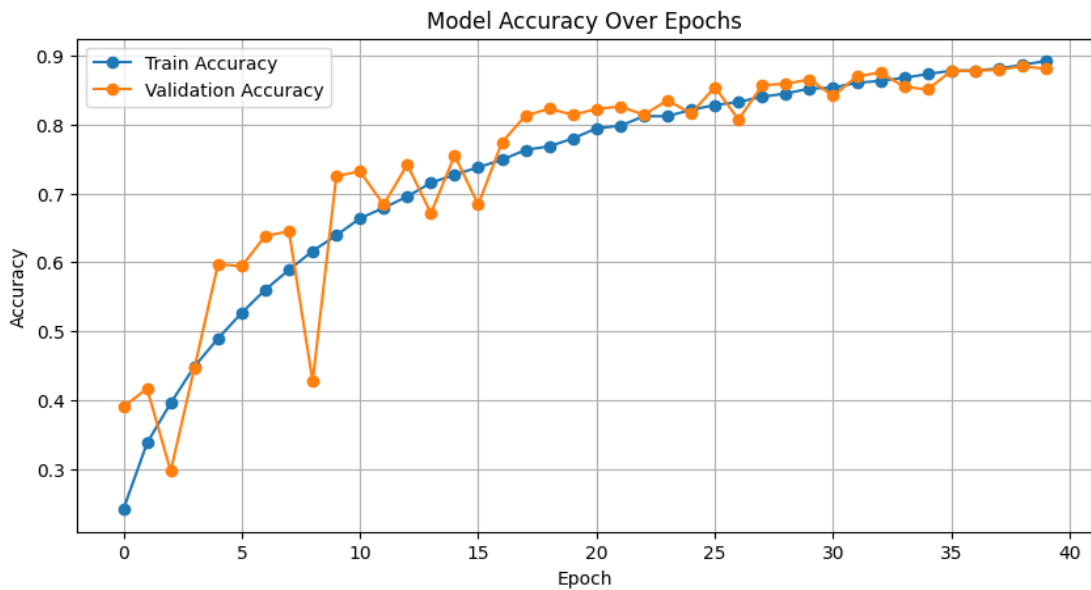


Figure 4.1: Training and validation Accuracy curve of ResNet-50.

The loss graph may further help in explaining the learning behavior of the same model. Figure 4.2 shows that both training and validation loss steadily declined throughout the training. It means that the classifier could minimise the error of prediction and enhance its belief in giving the right waste labels. The progressive drop in both the losses is also typically a symptom of balanced optimization and substantial feature extraction, which substantiates the applicability of the ResNet-50 architecture in the task.

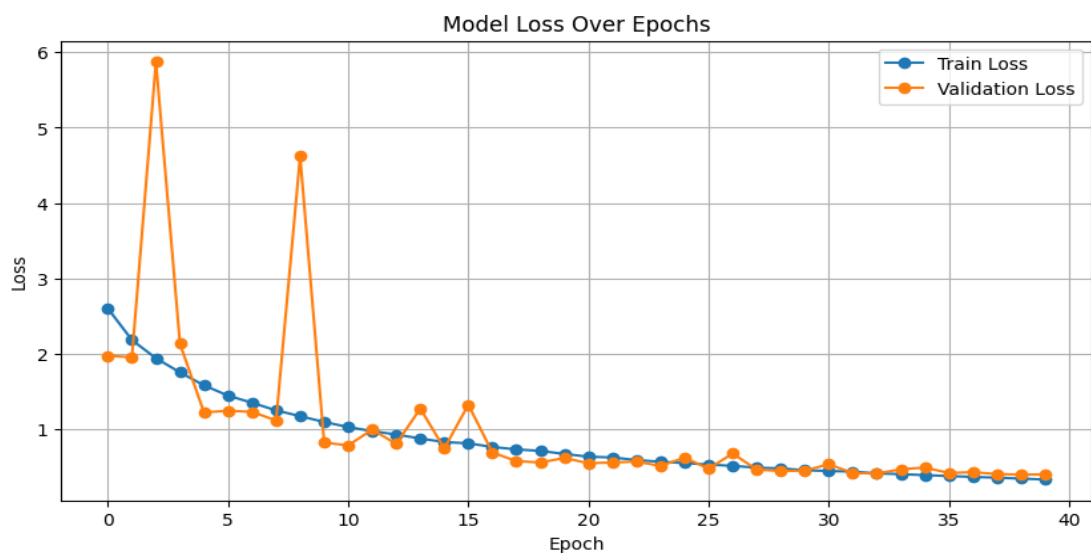


Figure 4.2: Training and validation loss curve of ResNet-50.

The confusion matrix gives a more in-depth picture of the performance by classes. Figure 4.3 indicates that the classifier worked very well on some classes compared to others which is a common occurrence in a multi-class waste recognition problem. F1-scores of 0.96 and 0.92 respectively indicate the best performance of clothes and shoes respectively. This means that the classes that could be more visually distinct were rated more reliably and similar classes created confusion. So, the confusion matrix is significant as it reflects the way in which the model performed on individual waste types rather than displaying the overall accuracy.

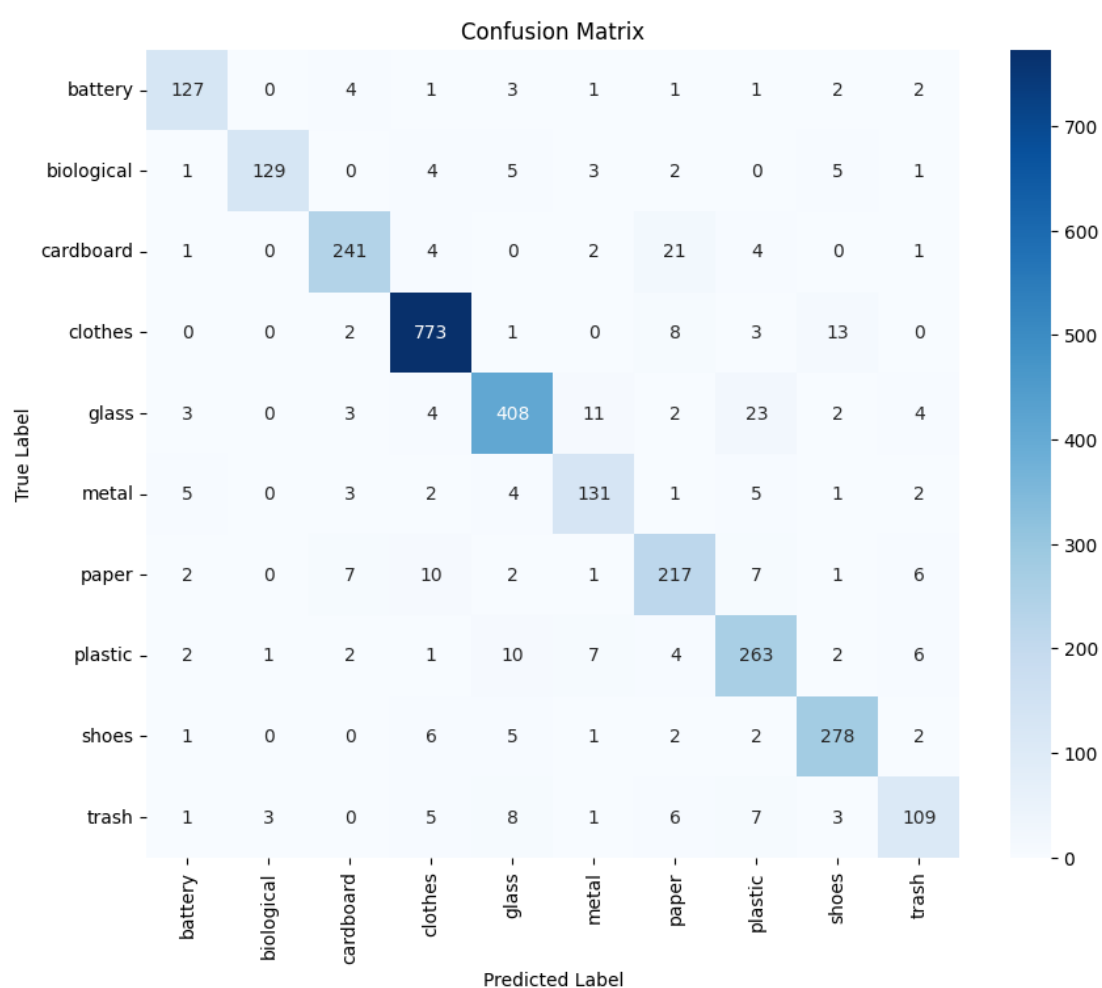


Figure 4.3: ResNet-50 Confusion Matrix.

Table 4.1 is a summary of the key findings of such a classification system. The last accuracy and the classes with the best performance are also pointed out in the table, thus, simplifying the interpretation of the results in a concise form. It is preferable to provide the results in a table following the discussion shown in a graph since it is a concise academic overview of the model performance.

Table 4.1: ResNet-50 Results with Rule-Based Recycling System.

Model	Resnet-50
Final Accuracy	90.83%
Precision	0.91
Recall	0.89
F1-score	0.90

Overall, the ResNet-50 system demonstrated that deep residual learning might be applicable to the automation of the waste classification process and to support the recycling decision mapping process with the help of the rule-based system. In the meantime, the results showed that the performance of classification could be further optimized with the assistance of more powerful feature-learning structures, which led to the development of hybrid models in the next step.

4.1.2 Comparison Analysis of Two Hybrid Models :-

Hybrid feature-learning architectures were investigated in order to enhance the classification performance even more. Two models were studied in this step: DenseNet121 + EfficientNet-B0 and ResNet-50 + Vision Transformer (ViT). The two models were also paired with a rule-based recycling system in such a way that the category of the waste could be directly related to a specific recycling process. The aim of this comparison was to find out which hybrid architecture would be able to represent features more strongly and more accurately classify waste.

One of the hybrid classification systems was the DenseNet121 + EfficientNet-B0 model. In this model, two CNN-based backbones were used in a way that compliments of visual features were learned on the same waste image. Figure 4.4 demonstrates the training and validation accuracy curves of this fused architecture with respect to epochs. The graph can be viewed as the fact that using two feature extractors was more effective in improving the quality of the representation and also learning classification in a stable way.

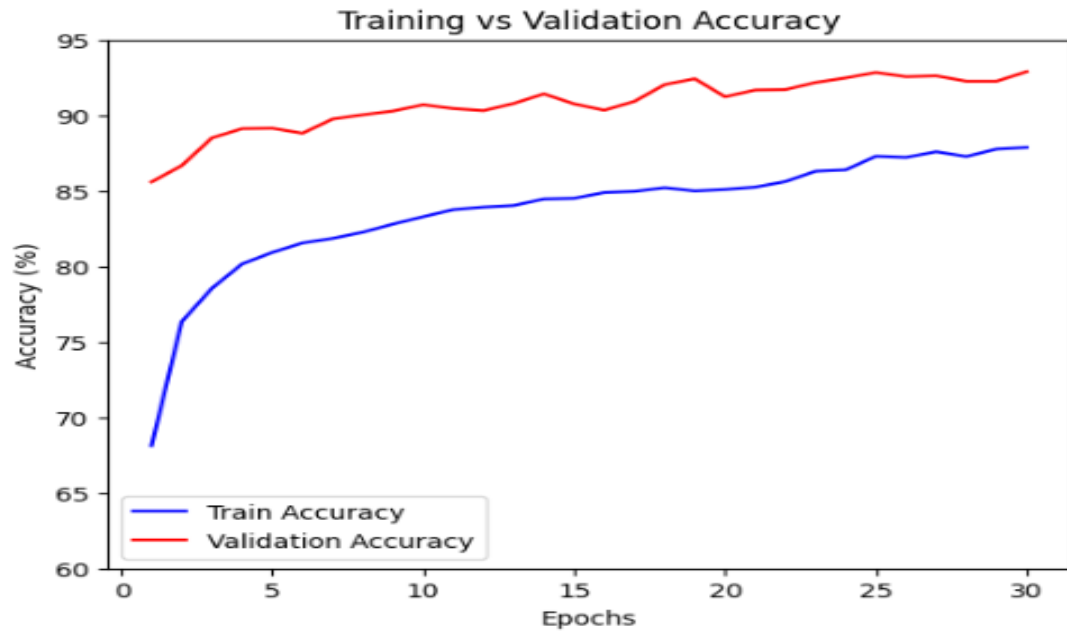


Figure 4.4: Training and Validation Accuracy of DenseNet121 + EfficientNet-B0.

The loss behavior of the same model is shown in Figure 4.5. The downward tendency of both training and validation loss suggests that the hybrid CNN structure could minimise the classification error, and could stabilise the feature space optimisation. This graph is significant as it is a complement of the accuracy graph and assists in the explanation that the increase in classification was justified by appropriate convergence in training.

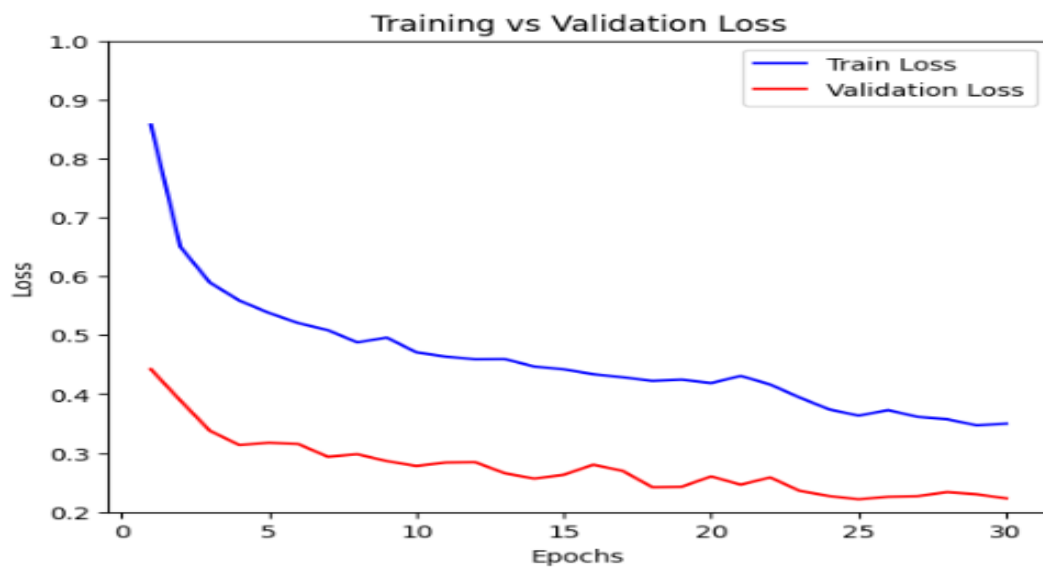


Figure 4.5: DenseNet121 + EfficientNet-B0 Training and Validation Loss Graph.

The confusion matrix gives a class-wise insight into the DenseNet121 + EfficientNet-B0 model. Based on Figure 4.6, the matrix indicated the correct recognition rate of each category and the presence of confusion between the visually similar materials. The number can be used to understand the behavior of the model at a single accuracy value and whether the hybrid CNN framework has enhanced the category-level prediction quality.

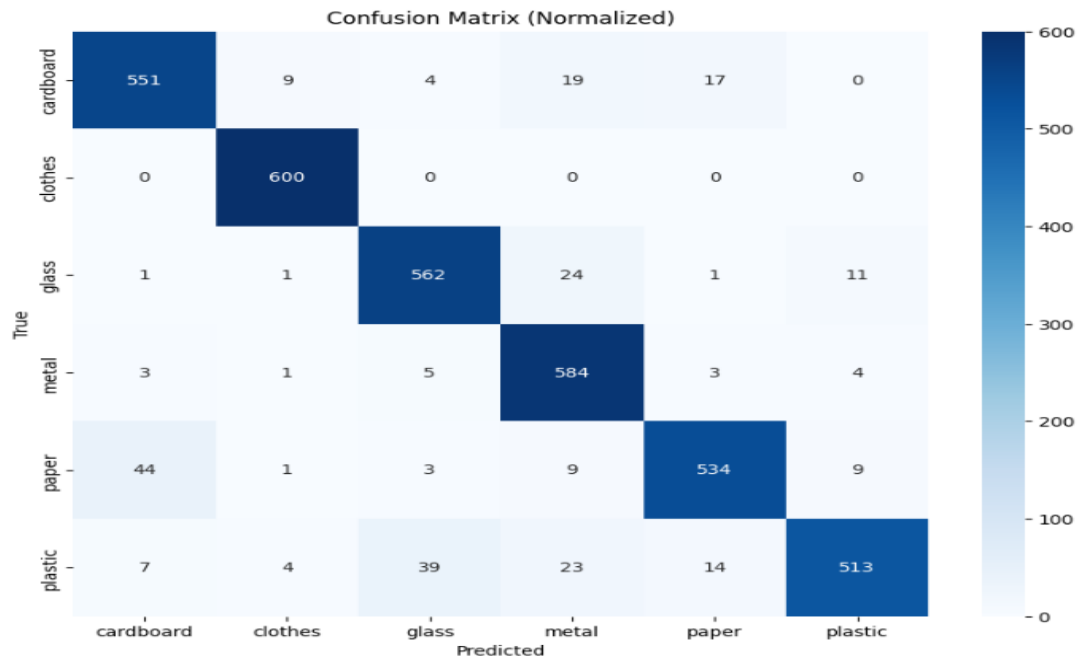


Figure 4.6: Confusion Matrix of DenseNet121 + EfficientNet-B0.

The combination of ResNet-50 and Vision Transformer were then combined to create a stronger hybrid architecture. This model combines local convolutional features extraction with global contextual learning thus rendering the classification system more expressive and powerful. Based on the received results, this hybrid model was able to obtain the final accuracy of 97.03% which is significantly greater than that of the 90.83% that was reached by the ResNet-50-only classifier. This gain is a clear indication that local texture-based learning coupled with transformer-based contextual understanding can significantly improve waste classification.

The results as presented in Figure 4.7 indicate that there was an increase in the training and validation accuracy of the ResNet-50 + ViT model with the number of training epochs. This value confirms that the hybrid architecture learnt successfully and had good validation performance implying good generalization. This better validation

accuracy is particularly significant as it demonstrates that the model did not merely memorize the training data, but also learnt transferable representations to unseen samples.

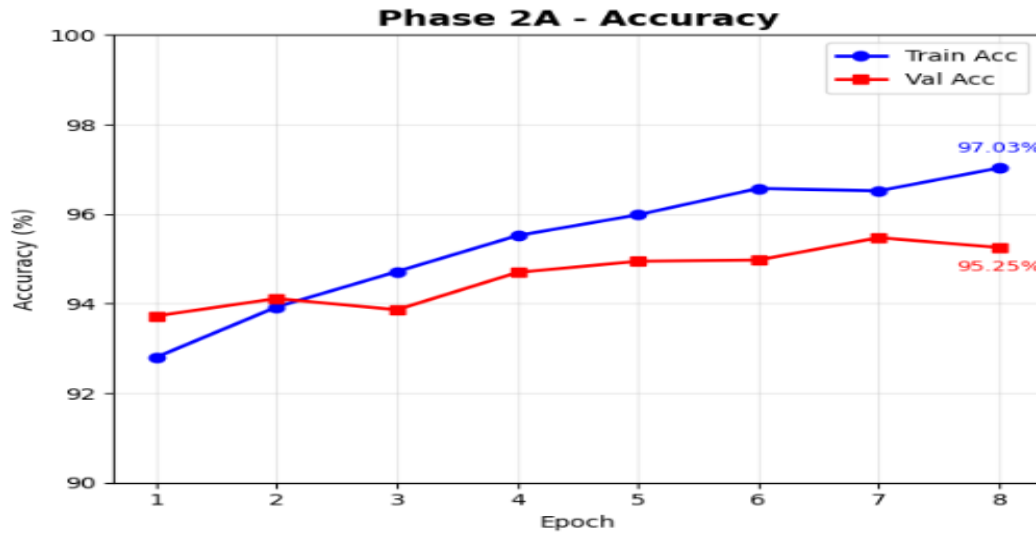


Figure 4.7: Training and validation accuracy graph ResNet-50 + ViT.

A loss behavior is shown in Figure 4.8. The decrease in loss over training shows that the hybrid model was effective in the reduction of classification error and converged in a steady-state. This helps to conclude that the CNN-transformer system was more effective to learn the visual representation of the waste classes in comparison with the previous classification systems.

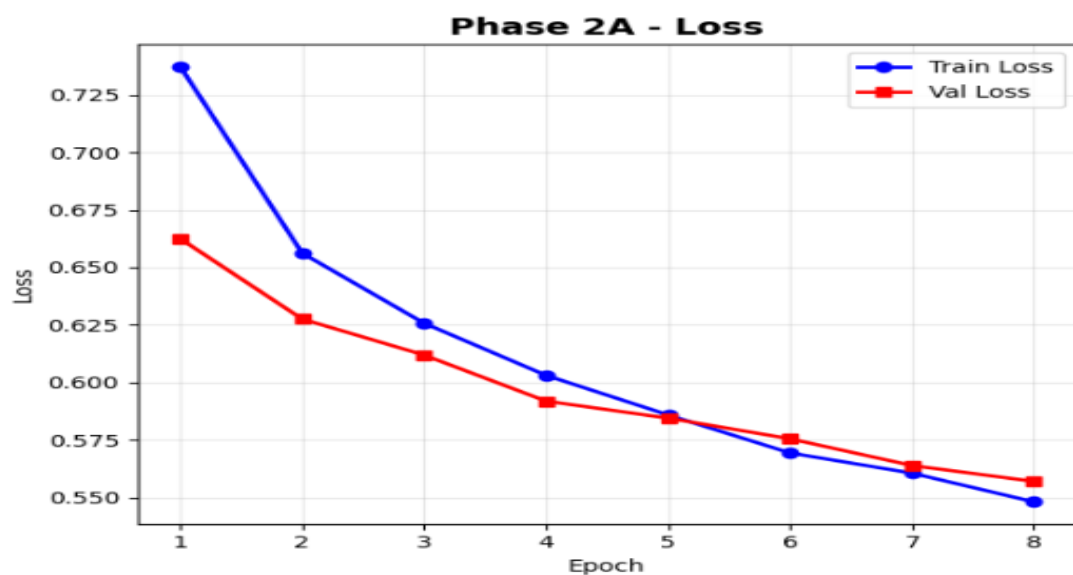


Figure 4.8: Training and validation loss graph of ResNet-50 + ViT.

The confusion matrix represents the effectiveness of this model in terms of the classes. As Figure 4.9 shows, the ResNet-50 + ViT model made more robust predictions in all the waste categories and had less confusion between classes compared to less complex classification systems. This value is significant, as it visually validates the argument that the hybrid architecture had an advantage in quality of recognition due to joint local and global feature learning.

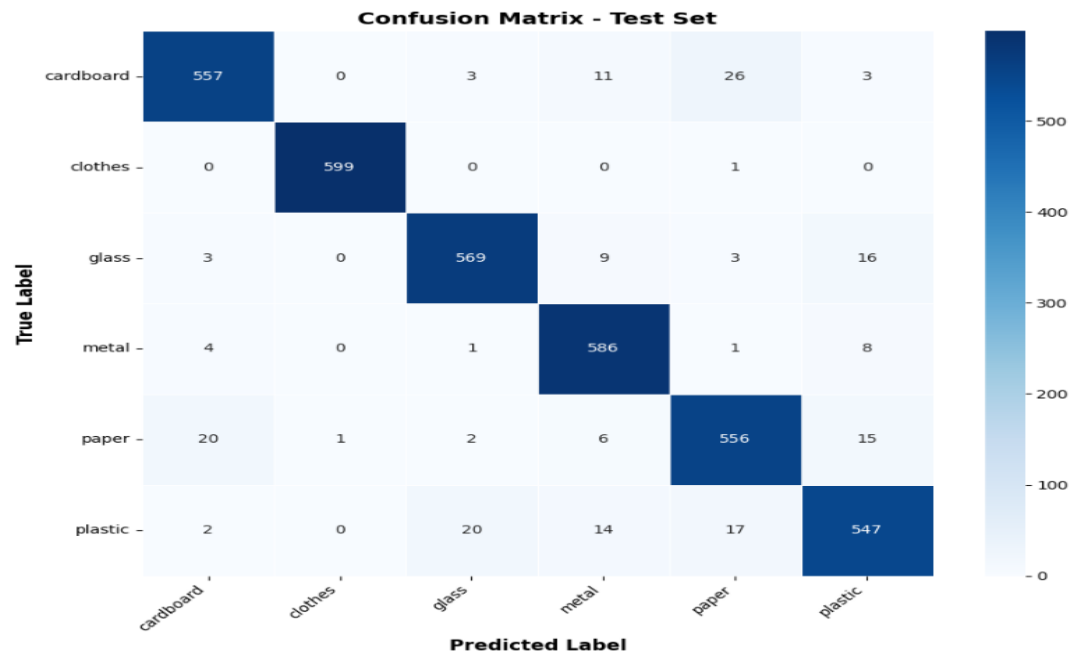


Figure 4.9: Confusion Matrix of ViT + ResNet-50.

The relative performance of the classification systems is summarised in Table 4.2. The table is helpful, as it gathers the key classification systems and shows the differences in their performance in a relative way. It makes it very clear that the ResNet-50 + ViT model gave the best classification-stage performance of the presented methods.

Table 4.2: Comparative Tables on Classification Models.

Metrics	VIT+ResNet-50 Hybrid	EfficientNet-B0 + DenseNet Hybrid
Accuracy	94.86%	92.35%
Precision	0.95	0.92
Recall	0.95	0.92
F1-Score	0.95	0.92
False Negatives	Very Low	Low
Generalization	Strong	Low-Moderate

In general, the hybrid classification phase evidenced that the combination of several feature-learning approaches results in more robust visual representations and waste classification. These findings also gave the needed background to expand the system to multi-object waste detection and contextual recycling advice in the ultimate system.

4.1.3 YOLOv8 and RAG Results :-

The last framework went past classification at the image level and proposed a Multi-object waste detector system based on YOLOv8 with Retrieval-Augmented Generation. This framework is more feasible as it is able to identify multiple waste items in a single image, localize them in a bounding box, and then provide a list of recycling recommendations depending on the materials found. This design would be more applicable to real-life situations of waste management where a variety of waste products can occur jointly and users need to be identified and guided to do something.

Table 4.3 will be the best way to introduce the overall performance of the detector. The values presented in the table show that the framework achieved Precision = 0.619, Recall = 0.451, F1-score = 0.522, mAP with threshold 50 = 0.535, and mAP with threshold 50-95 = 0.372. These measurements show that the detector could achieve meaningful learning behavior with respect to multi-class waste localization and

classification and stable practical performance. The strengths and limitations of the detector are also clearly summarized in the table prior to the discussion of the graph level analysis.

Table 4.3: YOLOv8 RAG Framework Performance Metrics.

Precision	0.619
Recall	0.451
F1-Score	0.522
mAP50	0.535
mAP50-95	0.372

The mAP curve is the most significant curve of object-detection performance. Figure 4.10 shows that the mAPs slowly increased throughout the training process, and the mAP at 50 reached about 0.535. This number proves the fact that the YOLOv8 detector has increasingly enhanced its capacity to localize and classify waste items throughout the epochs. A consistently rising mAP curve is a big signal that the object detector has acquired valuable visual and spatial representations based on the training data.

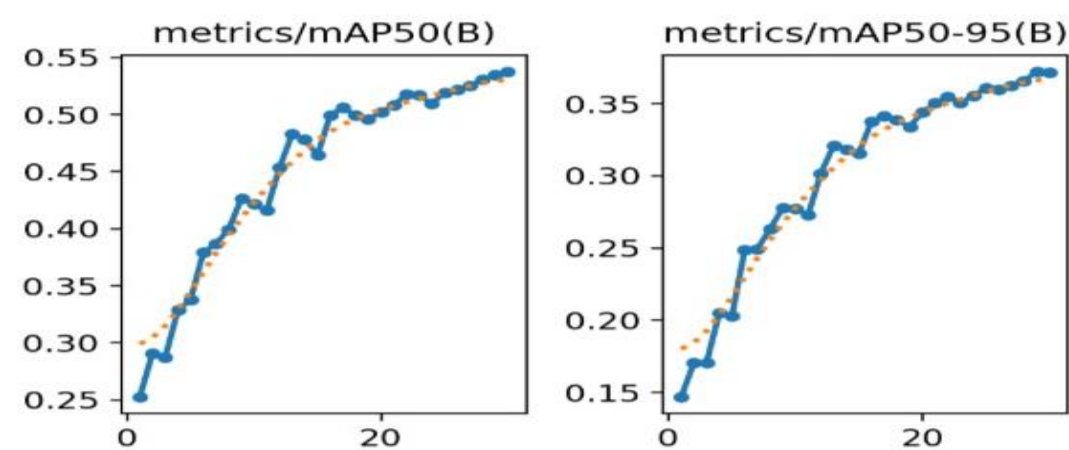


Figure 4.10: YOLOv8 mAP at 50 and 50-95 Graph.

The training box loss further expounds the localization ability of the detector. Figure 4.11 shows that the training box loss was reduced to 1.12 as compared to the previous training box loss of approximately 1.40, indicating that the model was slowly becoming more competent in the accurate placement of bounding boxes around waste objects.

This decrease in localization error suggests a significant increase in the quality of object-level prediction in training.

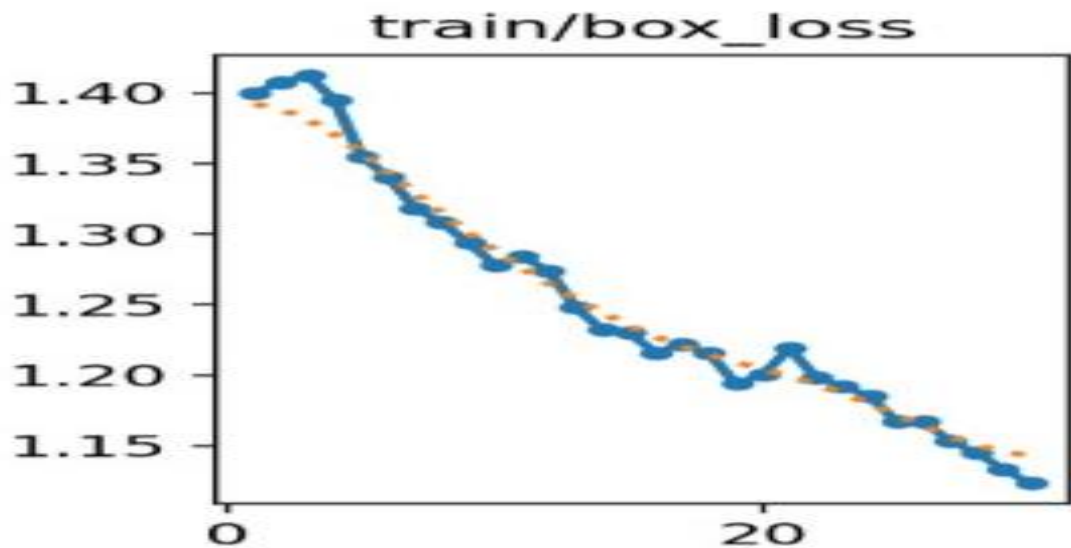


Figure 4.11: Training Box Loss Graph of YOLOv8.

The same is observed in the validation box loss. As observed in Figure 4.12, the loss on the validation box reduced to about 1.16 as compared to the 1.52 and this means that the patterns on which localization patterns have been learned are also able to generalize to the unseen validation data. What made this graph significant is that it indicates that the detector not only fitted the training set but did not drastically change its learning behavior as it would have done with severe overfitting.

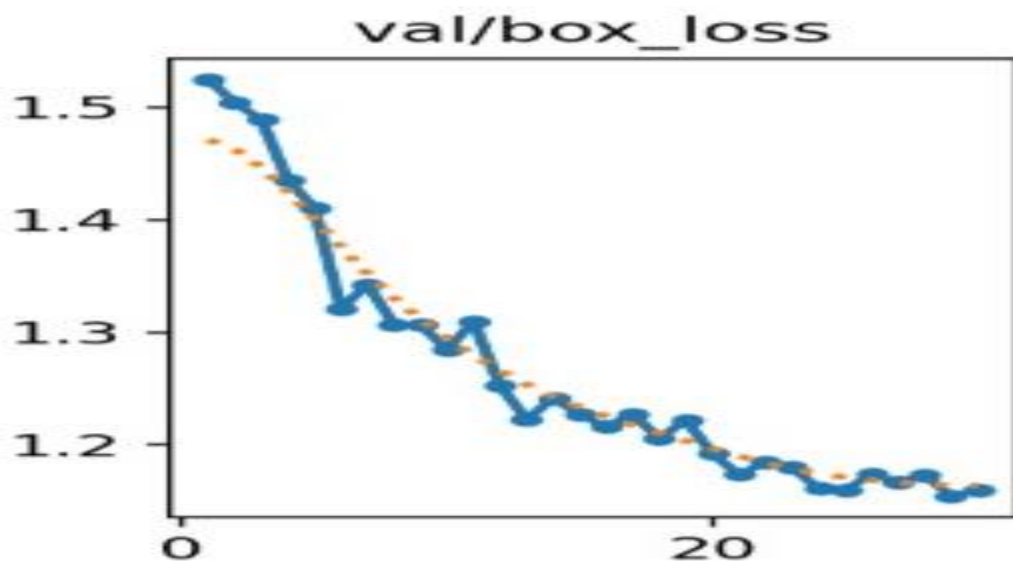


Figure 4.12: Box Loss Graph of YOLOv8-Validation Box Loss Graph.

The F1-confidence curve gives more information as to the balanced performance of the detector at various confidence thresholds. Figure 4.13 indicates that the total F1-score was approximately 0.51 at a confidence of 0.381, indicating the most balanced operating point of the detector. The analysis of the classes of curves further reveals that the glass was the best, then metal, biodegradable waste and cardboard, with paper being the most challenging. This number is quite handy as it not only points at the optimal working threshold of the detector but also the difference in the quality of the detection on the class-level.

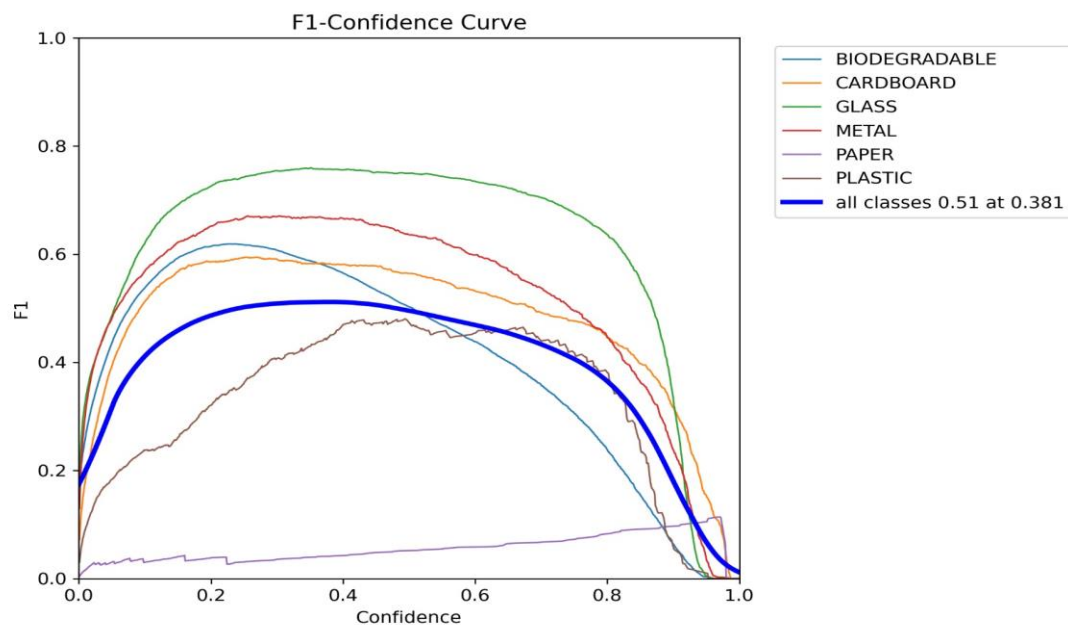


Figure 4.13: F1-Confidence Curve of YOLOv8.

The confusion matrix is also used to explain the class-wise prediction distribution. The detector as seen in Figure 4.14 had a higher recognition of glass and metal and paper had a lower correct-class concentration and increased confusion. A certain degree of confusion with the background can be seen as well, which can justify the mediocre recall value. The confusion matrix is thus significant in the connection between the general metrics and the real class-wise prediction of the model.

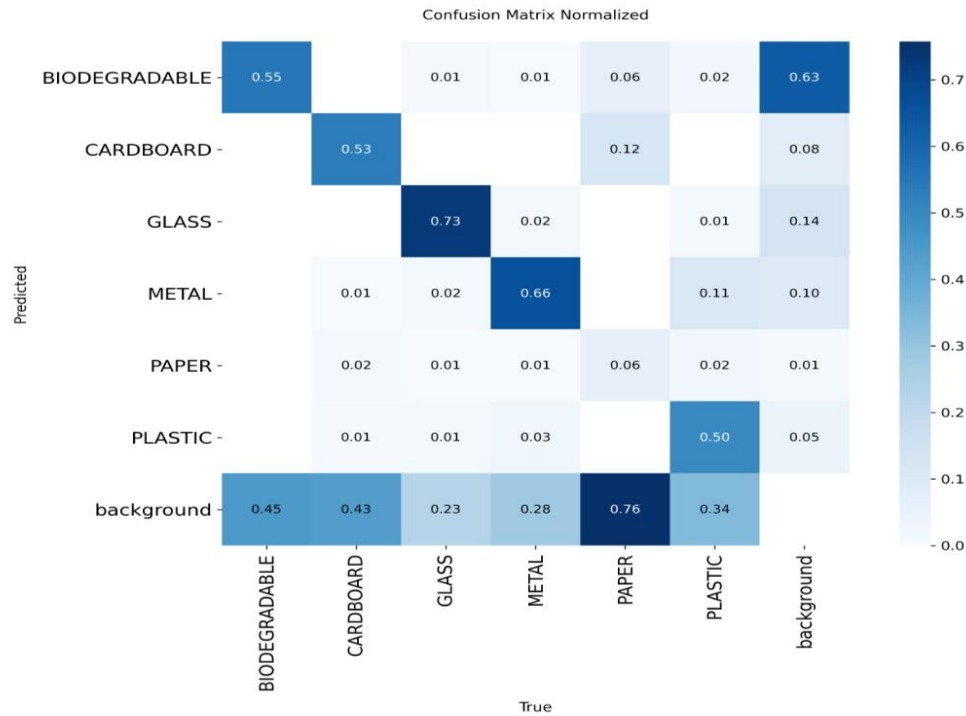


Figure 4.14 Confusion Matrix of YOLOv8.

In general, the YOLOv8 and RAG architecture showed consistent multi-class object-detection results and was able to scale the system to a viable intelligent waste management system. In contrast to the previous classification systems, this framework can be used to assist with multi-object recognition, localization based on confidence, and category-specific recycling instructions. Thus, it is the most comprehensive and pragmatic product of the whole work.

4.2 Test case results/Result Analysis :-

This section describes how the developed systems behave in practice with respect to the relationship between the input provided to the model and the output provided by the system. Although the above section has dealt primarily with graphs, accuracy trends, losses and confusion matrices, the following section is aimed at describing how the systems respond to real-life testing. That is, this section demonstrates the type of image that is inputted into the model, how the model processes that image and the type of output that it produces. Such an explanation plays a crucial role in a thesis as it links the numerical findings to the real working behavior of the developed framework.

4.2.1 ResNet-50 with Rule-Based System Results :-

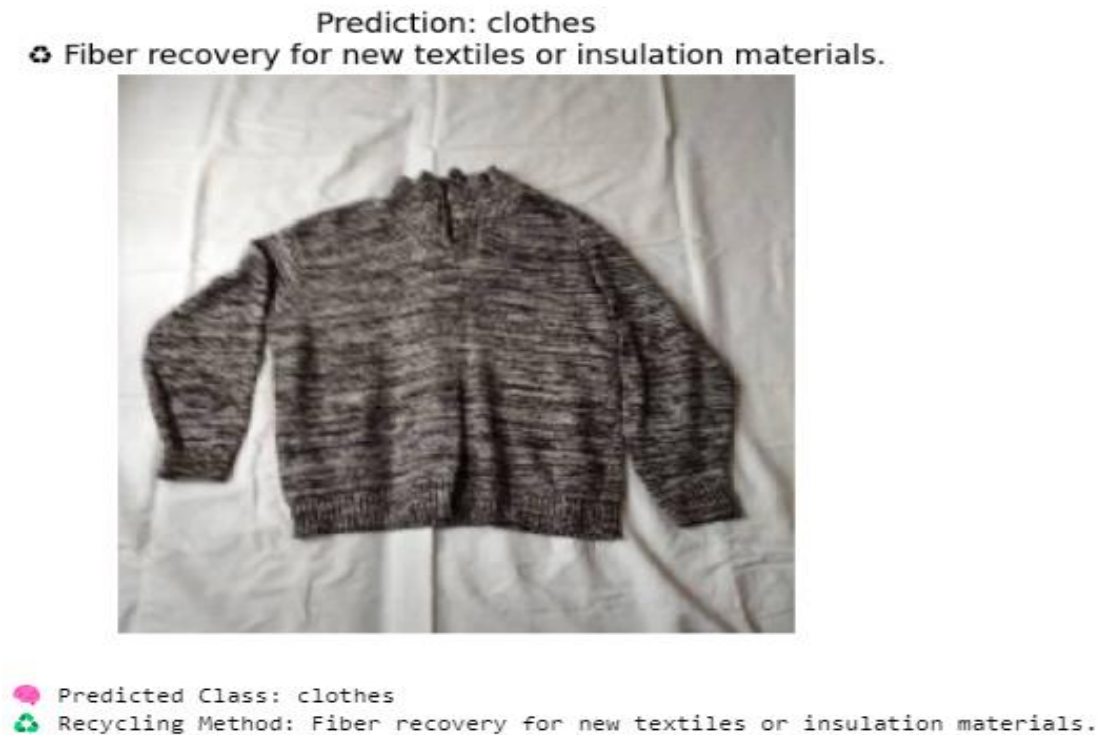


Figure 4.15 Shows the results with the rule-based system using ResNet-50.

In the ResNet-50-based system, the test case starts with an input of a waste image that is presented to the classifier. This model is an analysis of the visual contents of the image and predicts the most probable waste category out of the supported classes. After the class is known, the rule based mechanism corresponds that class to a suitable recycling method. Thus, the system output does not consist solely of classification, but also of a particular recycling decision, depending on the type of the predicted waste. This renders the testing result more viable since the user is given the recognition and direction of an input.

The input image to the classifier is the first step in the test case as indicated in Figure 4.16. This number must show the actual waste picture prior to any forecasting, which will enable the reader to clearly see what sort of visual information is being run through the model.



Figure 4.16: Input Image to ResNet-50 Classification.

After the image is processed by the model, the system generates the predicted waste category. The output (shown in Figure 4.17) must show the type of waste that has been classified and the recycling method generated by the rule-based system. To illustrate this, when the model recognizes the item of waste as plastic, the system output must reflect the plastic label and the associated recycling activity that is related to that type



Figure 4.17: The Result of Predicted Waste Class and Rule-Based Recycling Result of ResNet-50.

The analysis of the results of this test case demonstrates that the framework is able to convert a raw waste image into a meaningful output of classification and subsequently an action related to recycling. This is significant as it shows the usefulness of a combination of a deep learning-based classifier with a rule-based decision system. The final accuracy of the model was 90.83; therefore, the output of the system could be taken as a fairly reliable one to show the automated waste classification and disposal support. The described class-based strength of clothes and shoes, where F1-scores are 0.96 and 0.92, once again suggests that classes that visual differentiation is more prone to produce correct outputs when tested in practice.

In general, the test case outcomes validate that the ResNet-50 using the rule-based system is capable of taking a waste image as input, recognizing its type, and providing a valuable output related to recycling. This confirms the empirical performance of the system and demonstrates that it can be used as an automated aid to waste classification.

4.2.2 Hybrid Classification System Results :-

The hybrid classification step was invented to enhance the performance of waste recognition by integrating complementary deep learning feature extractors. Two hybrid systems were taken into consideration in this stage. The former used a combination of DenseNet121 and EfficientNet-B0, and the latter used a combination of ResNet-50 and Vision Transformer. In both examples, a waste image is given to the system, which is then predicted to be of a specific type of waste, and the predicted type of waste is then mapped to a recycling process by a rule-based system. These systems have a similar practical testing behavior to the previous classifier, however, the quality of the prediction should be better due to more effective feature extraction.

The initial hybrid test case starts with the DenseNet121 + EfficientNet-B0 model. The input to the system is a waste image as shown in Figure 4.18 that is fed to the hybrid CNN-based classifier. This figure must show the raw sample image before classification to ensure that the reader is able to see the material appearance through which the model is analysing.



Figure 4.18: Input Image to DenseNet121 + EfficientNetB0 Model.

Once the image has been processed, the classifier produces the forecasted waste category. Figure 4.19 shows that the output must show the final waste label created by the model and the resulting recycling output created by the rule-based mechanism. This result is significant as it shows that the hybrid CNN architecture does not only enhance the visual features learning, but also allows practical decision making in recycling based on the material predicted.



Figure 4.19: The output of DenseNet121 + EfficientNet-B0 with Rule-Based Recycling Result.

The second test example is about the ResNet-50 + Vision Transformer model. In this model, the waste image is once more the input, but the prediction is produced by a blend of the local convolutional feature extraction and the global contextual understanding. The input image must be presented prior to classification as indicated in Figure 4.20 to enable the reader to see the waste sample as it is fed to the hybrid CNN-transformer architecture.



Figure 4.20: The Input Image to ResNet-50 + Vision Transformer Model.

The production of this more powerful hybrid model must then be presented in Figure 4.21. It should be represented by the forecasted waste category and the corresponding recycling advice produced by the rule based system. As the final accuracy of this model was 97.03, the result, which this framework delivers, can be viewed as more credible and more consistent than the previous classification frameworks. Thus, the figure is significant as it shows that the better the hybrid feature representation is, the more viable is the performance of classification in testing.

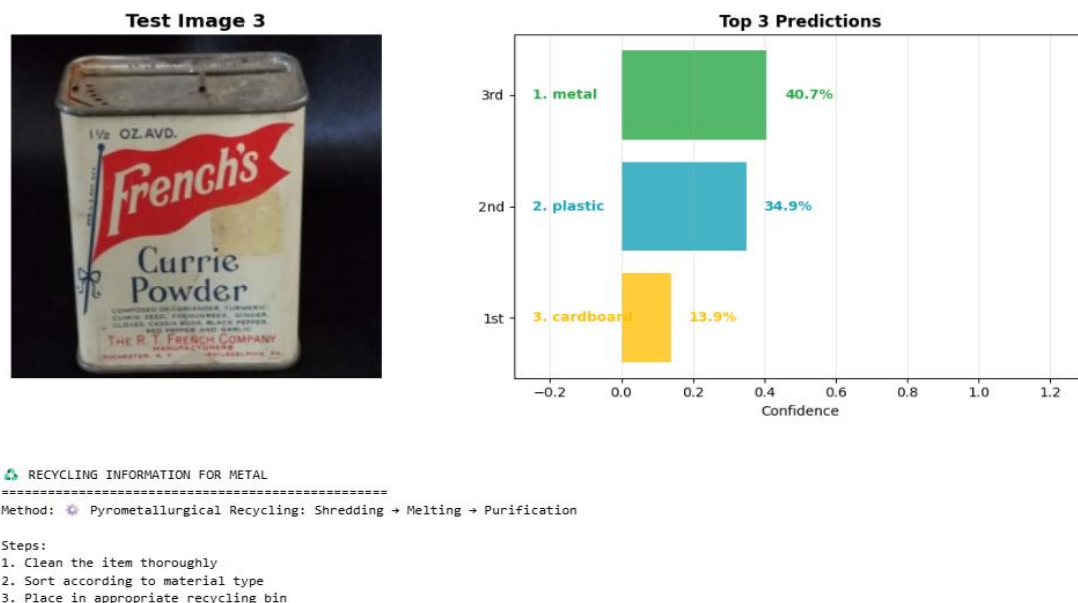


Figure 4.21: Result of ResNet-50 + Vision Transformer + Rule-Based Recycling Result.

4.2.3 YOLOv8 and Retrieval-Augmented Generation Results :-

The last framework contrasts greatly with the previous classification systems as it is not classified on the image level but on the multi-object garbage detection. Under this model, the test case starts with the user uploading an image that can have either a single or a number of waste materials. YOLOv8 model takes the image, detects all waste objects visible in it, wraps them with the bounding boxes, labels them, and gives them the confidence values. The identified labels are then sent to the Retrieval-Augmented Generation part, which produces contextual recycling advice depending on the recognized materials. Consequently, the products of this structure are more comprehensive, informative and practical compared to the products of the previous classification systems.

The uploaded waste image is the first step of the testing process as indicated in Figure 4.22. This figure is to show the original input image prior to detection to enable a reader see the mixed wastes scene provided to the framework. This is in contrast to the previous classification systems which have only one waste object per image.



Figure 4.22: YOLOv8 Detection Uploaded Input Image.

After processing the image, the detector generates the visual recognition output. Figure 4.23 displays the predicted waste and labels as well as confidence scores in the same image as bounding boxes, which should be displayed as output. The significance of this figure is that it shows that the system is capable of localizing and classifying a number

of waste objects at the same time, instead of just classifying the entire image under one classification. It is thus the key visual identification of the framework.



Figure 4.23: Image of Results with Bounding Boxes, Waste Labels and confidence score.

After the detection step is finished, the labels of the detected waste are sent to the retrieval and generation pipeline. The resultant output should be the recycling recommendation according to the discerned types of waste as illustrated in Figure 4.24. This output may be in form of material specific instructions of cleaning, sorting, separation, recycling or proper disposal. This figure is also significant as it indicates that the framework does not just involve visual recognition but also offers the user the contextual decision support. That is, the output stage converts the identified types of waste into valuable and practical environmental information to the user. This enhances the usefulness of the system in real life, as the user does not get the name of the waste material detected. Rather, the framework describes the second step that needs to be taken to handle waste in a proper way. This output may be significant in the real-world waste management, in which proper disposal choices are just as significant as proper waste identification. Thus, this step helps to prove that the framework links computer vision with knowledge-based aid in a purposeful and user-friendly way.

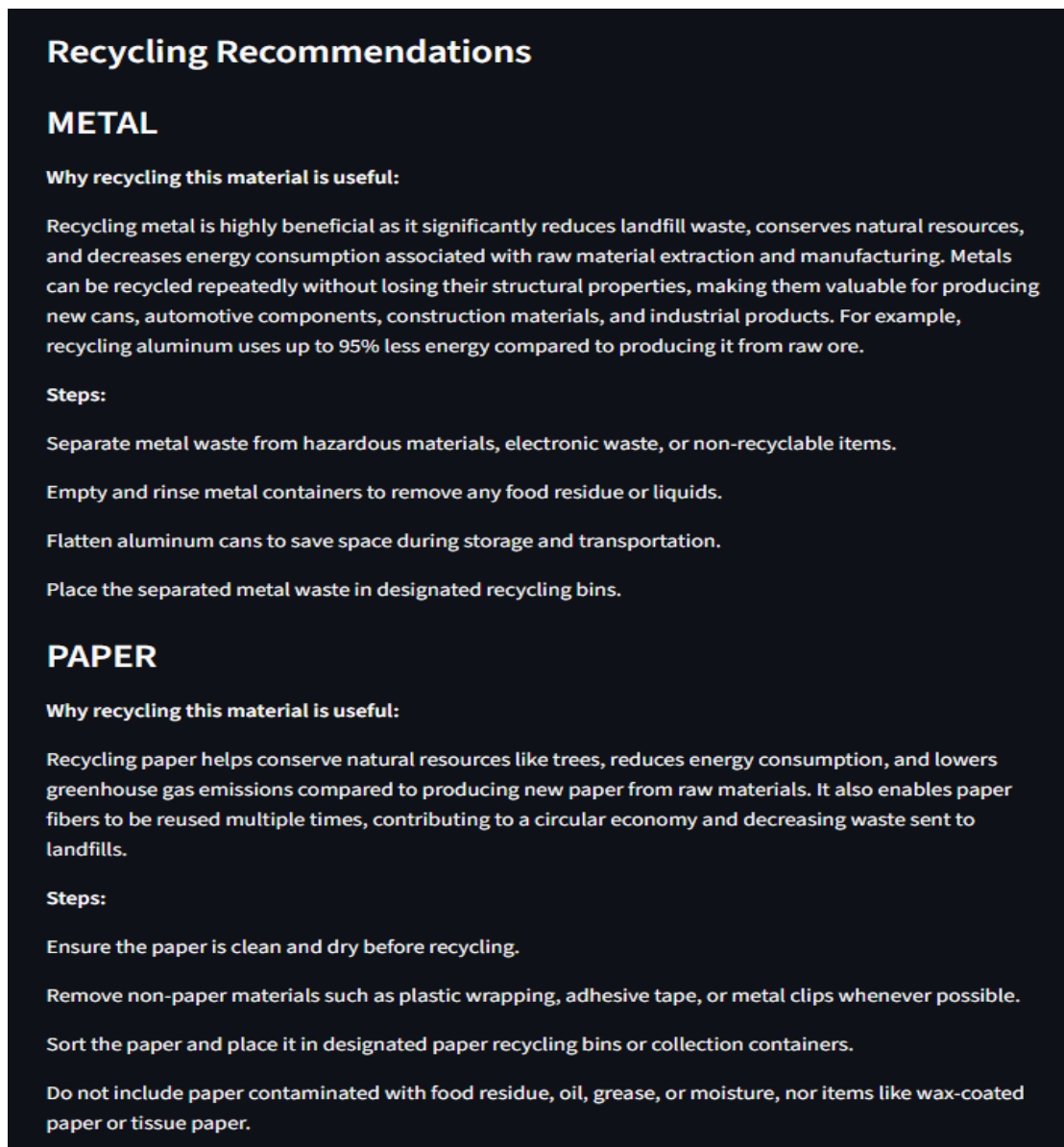


Figure 4.24: Recycling Guidance Generated by Recycling Detected Waste Classes.

Application interface is the most descriptive way of expressing the entire working behavior of the framework. The interface is to display the uploaded image, the results of the detected waste, and the generated recycling recommendations in a single view as seen in Figure 4.25. This is a meaningful number since it indicates usability of the system end-to-end and verifies that the framework has been used as a working application and not merely as a model-level experiment.

Multi-Class Garbage Detection with Recycling Recommendation

Upload a waste image to detect waste classes and get recycling guidance.

Upload an image



Drag and drop file here

Limit 200MB per file • JPG, JPEG, PNG

Browse files



istockphoto-907950256-612x612.jpg 74.6KB



Confidence Threshold

0.25



Uploaded Image



Figure 4.25: User Interface with Detection and Recycling Guidance.

The analysis of the results of this framework indicates that this framework is more practical and more informative than the previous systems of classification on an image level. The former systems generated only one waste image, which in turn led to one predicted class and one possible recycling action. Within the current system, one is able to handle mixed waste sites, detect various types of waste, visualize their location, visualize confidence scores, and provide contextual recycling advice. In such a way, the output will be more comprehensive, more realistic, and more appropriate to practice-based waste management support.

These practical outputs are also effective as evidenced by the detection metrics that have been reported above. The framework achieved Precision = 0.619, Recall = 0.451, F1-score = 0.522, mAP with threshold 50 = 0.535, and mAP with threshold 50-95 = 0.372, which indicates stable multi-class detection performance. These values prove the fact that such visual products displayed in the figures are justified by the presence of significant detection power and are not just some random cases.

In general, the results of the test case and analysis of the results indicate that the final framework is effective to convert a raw waste image into a full series of outputs, i.e., detection, localization, classification, and recycling recommendation. The framework is more applicable in the practical application of sustainable waste management in the real world than the previous classification-based systems.

4.3 Observations from the Work

As the observations made on the whole piece of work make it clear, the system was not only improved regarding technical performance, but also regarding the practical usefulness, interpretation of the results, and the applicability of the results to the actual waste management issues. As the various developmental levels progressed the work was increasingly refined to less waste-classification at the level of images and more to hybrid feature-learning methods and eventually to an all-out multi-object waste detection and recycling guidance system. This gradual development in itself is one of the most significant observations, owing to the fact that it demonstrates that the issue of intelligent waste management cannot be resolved completely with the help of classification only. Rather, it involves the integration of the correct visual identification, improved feature depiction, useful output interpretation and lastly recycling contextual support.

Among the initial significant findings of the work, it is important to note that deep learning-based waste classification proves to be very effective in comparison with manual and traditional methods of identifying waste. The ResNet-50-based classification system has shown that a deep residual architecture can learn meaningful visual patterns on waste images, and can attain high baseline classification accuracy. The final accuracy of 90.83% obtained shows that the model was able to extract discriminative features of the data and place the appropriate waste labels on most of

them. It is a significant finding as it proves that a single powerful convolutional model can automate a large portion of the waste classification procedure and minimize the use of manual classification.

The other observation closely related is that various types of wastes do not affect the overall classification performance in similar ways. Each of the classes demonstrates that visually distinct classes like clothes and shoes performed especially well with F1-scores of 0.96 and 0.92, respectively. This implies that classes whose texture pattern is more identifiable, the form more identifiable, and those with more identifiable appearance to the eye will be classified by the model at a higher rate. Conversely this also means that likely there would be more confusion in classes that overlap with each other in terms of texture, shape or color. Hence, a key learning during the work is that the performance of waste classification is not only influenced by the quality of models, but also the visual distinguishability of target classes.

The initial classification framework learning curves are also helpful in making observations. The gradual increase in the training as well as the validation accuracy along with the gradual decrease in the training and validation loss suggest that the model was learned in a stable and consistent way. This demonstrates that the choice of preprocessing steps, training process, and network structure fitted the waste data. More to the point, it appears that the training and validation curves are very close, which indicates that the model has reasoned generalization rather than just memorization of the training samples. This is a significant finding since a model that is to be used in a real-world waste analysis should not only be effective in perceived images, but also in unexplored and unperceived waste samples.

Another conclusion of the initial findings is that confusion matrices are incredibly useful in waste-classification research. The overall accuracy alone is only a general summary of the performance but the confusion matrix would show what categories to be predicted correctly and in which the model is prone to errors. Working on the work, it became obvious that the analysis in terms of classes is required as the issues of waste recognition may concern material, which is visually similar. In this way, a significant lesson is that waste classification systems must be always reviewed on aggregate metrics and on the distribution analysis on classes. It makes the interpretation more significant and assists to understand what classes should be further improved.

One of the most significant findings of the later classification stage is that hybrid type of architectures are more effective when compared to single backbone classifier where more extensive feature learning is required. The hybrids discussed in the paper revealed that the waste recognition performance is enhanced when a variety of feature extraction strategies are combined. Specifically, the hybrid architecture that united ResNet-50 with a Vision transformer reached a final classification accuracy of 97.03 which is significantly greater than the 90.83% that the previous ResNet-50-only model reached. The importance of this performance difference is very high since it shows that the quality of classification is better in case local spatial detail is acquired with global contextual understanding.

This brings to the next important point that local visual information may not always be enough to achieve extremely reliable waste classification. The conventional convolutional models excel at capturing edges, textures, and material patterns, but waste images may have contextual dependencies that cannot be best represented by local convolution, alone. The increased performance of CNN- transformer hybrid suggests that relations at the image level (global level) and at the patch level (context) contribute significantly to the correct recognition of waste materials as well. Thus, among the key findings of the work is that the classification of waste can be advantageous when based on architectures with both local and global representation learning as opposed to relying on a single visual learning strategy.

The comparative analysis of hybrid models also indicates that feature fusion is a valuable design option to the waste analysis system. The system can integrate the outputs of various models thus being able to capture a broader spectrum of visual information. This not only increases the level of numerical accuracy, but also increases the practical reliability of the output of the classification. In practical application, it will result in the system having a higher probability of generating an accurate waste label when the appearance of the material is complicated, partly unclear, or similar to that of another category. Thus, the next insight of the work is that the more diverse a feature extraction is in architecture, the more powerful and stable the classification behavior is.

Simultaneously, the classification-stage findings demonstrated that there is a significant weakness: even highly accurate image-level classifiers cannot be completely relied upon in the practice of waste management. This is due to the fact that image

classification gives one label to the whole image when in reality, waste scenes in the real world have a number of waste materials in a single frame. This constraint came to be one of the most significant observations made in the work. It demonstrated that a model can be very accurate in classification but it can still be practically restricted in case it is not able to localize many items in a mixed waste scenario. That is, good performance in classification does not necessarily lead to performance in the real world when the real task is in object-level recognition.

This fact was the main reason as to why a shift towards an object detection framework was necessary. The last detection system, which relies on YOLOv8, showed that the transition to object-level recognition at image-level classification is much more effective in terms of practical applicability of the research. In contrast to the previous systems, the detection framework could detect several waste materials on a single image, encircle the waste materials with a bounding box, and provide a confidence score to each object detected. This is a significant finding since it signifies the departure of simplified waste analysis towards the more realistic waste management environment. In real life the structure was far more akin to the actual presentation of waste in real life, whereby several objects are combined together as opposed to being separated.

The qualitative findings of the detection framework also give some more observations. The model got metrics Precision = 0.619, Recall = 0.451, F1-score = 0.522, mAP with threshold 50 = 0.535, and mAP with threshold 50-95 = 0.372. These values demonstrate that the detector has actually learned to recognize meaningful multi-class objects-behaviour, despite the task being more challenging than image-level classification. An extremely significant point to note in this case is that the decline in results of accuracy-like tests in comparison to the results of classification cannot be perceived as simply a poorer performance. Rather, it should be perceived within the framework of difficulty of tasks. The model is not only required to classify but also to localize the object in question, which is not an easy task to undertake in the first place as compared to applying a single label to the overall image. So even moderate detection measures can be a useful and meaningful practical performance in a multi-object environment.

The results of the detection-stage also indicate that certain types of waste can be identified more easily than others. The analysis of the F1-confidence and discussion of the confusion-matrix indicates that glass and metal were the best, and paper was a more

challenging category. This observation is very significant as it indicates that the stronger the visual distinctiveness of the classes, e.g., reflection, shape rigidity or material texture, the easier it is to detect and localize the classes by the detector. Conversely, less visually definite or flatter materials have a higher chance of being mixed up with the background, or even with similar classes. It implies the data diversity, annotation quality, and balancing of the classes are particularly significant in the scenario of visually weak or confusing waste classes.

The other observation based on the detection results is that the training process was consistent during the learning process. According to the review presentation and the paper, the training box loss decreased to about 1.12 as compared to 1.40 and the validation box loss decreased to about 1.16 compared to 1.52. On the same note, the mAP50 improved steadily throughout the course of training and peaked at about 0.535 in the last stage. These trends show that the model learnt gradually and became better and better able to both localize and classify waste objects as time went by. This stability is noteworthy as it proves that the selected detection architecture and training configuration was suitable to the dilemma although the task as such was more complicated than the previous classification systems.

Another useful observation can be made on the F1-confidence graph. The paper observes that the total F1-score at a confidence level of 0.381 which is the most balanced operating point of the detector was approximately 0.51. This implies that the level of performance of the models is highly dependent on the selection of the detection confidence threshold. When the threshold is too low, false positives can be high and when it is too high, then true detections will be missed. Hence, a key practical point is that based on the application required, the operating point of the detector should be selected with care. This is particularly important in real world waste management systems where costs of false detections and missed detections can vary based on the situation.

One of the most important insights of the general work is that the framework proved to be much more helpful when waste recognition was incorporated with recycling guidance. The output of the previous systems was largely a predicted type of waste and a recycling route as mapped out by predetermined rules. The final system, though, enriched the output and made it more contextual since the waste labels detected were

sent to a retrieval and generation pipeline. This implies that the system was no longer recognized but it was a decision-support tool. This is one of the most significant observations made in the thesis since it demonstrates that visual intelligence is not enough to have practical waste management. The users must also be provided with instructions on what to do once the waste has been detected.

This brings to the next practical observation; the last framework has outputs which are more aligned to the real user needs. Users in the real-world waste management environment lack pure knowledge that an object is plastic, paper, metal or glass. They also would like to know whether it needs to be cleaned, separated, sorted differently, or recycled in a certain way or to be disposed of differently. The framework provides practical output, as it produces recycling guidelines once detected, instead of the abstract prediction. Thus, one of the most interesting findings is that the incorporation of contextual recommendation highly enhances the value and applicability of the created system.

Another good observation is made at the interface-level implementation. The last structure was not confined to training and assessment as a back-end model, but it was introduced in the form of an application interface where one can upload an image and view the identified waste products, and get a recycle recommendation. This is significant as it shows that not only is the developed work algorithmically valid, but it is also operationally usable. In most academic work, there is a model performance which is reported without practical interaction. In this piece, though, the combination of detection, creation of guidance, and presentation of an interface demonstrates that the framework has been created as a working end to end system.

The other thing that can be noticed is that the work displays a definite enhancement of the environmental relevance as the framework became more sophisticated. The first phase primarily dealt with automated classification of waste. The subsequent phases, though, helped to achieve more realistic segregation of waste, enhanced quality of recognition, and recycling guidance. This implies that the end product is technically superior in addition to being more meaningful to the environment. It helps handle wastes correctly, enhances the understanding of disposal practices and has a greater contribution towards the practice of sustainable waste management.

Another aspect to observe is the connection between performance metrics and practical significance. The best numbers in terms of metrics in this paper were obtained through the hybrid model during the classification stage; however, the most practical model that we have in this project is YOLOv8+RAG. Thus, we see that higher performance metrics do not necessarily mean practicality. Sometimes it makes more sense to use a somewhat lower performing system that seems more reasonable and more applicable to real-world situations. This thesis level point indicates that evaluation should take into account realism and relevance of the task in addition to accuracy.

Chapter 5

Conclusion and Future work

5.1 Conclusion :-

The contribution to the work in this thesis was aimed at creating an intelligent waste management system that would enhance the effectiveness of garbage recognition and recycling assistance with the help of artificial intelligence. The general aim was to minimize the drawbacks of the conventional waste management systems where manual sorting is usually slow, labour-intensive, and subject to inconsistencies. The work was gradually developed, through the various development phases, to higher levels of hybrid learning architectures, and eventually to a more realistic framework, of multi-object waste detection and retrieval-based recycling guidance. This advancement allowed not only to solve the problem of waste recognition, but also the problem of giving meaningful disposal and recycling assistance after recognition, which is also important.

The stages of the work, which were classification-oriented, proved that deep learning is very effective in automated waste recognition. ResNet-50-based system provided a solid baseline demonstrating that deep residual learning can effectively classify waste images and aid in recycling choices through a rule based system. This was enhanced by the later hybrid architectures which combined complementary feature-learning methods. Specifically, the hybrid model that combined ResNet-50 with Vision Transformer demonstrated a better classification accuracy that validated that local visual feature extraction with global contextual learning enhances the quality of waste classification. These findings demonstrated that hybrid deep learning has a greater representation strength than single classification backbone and is well-suited in the analysis of a complex waste-image.

The last phase of the work was to extend the system to levels that do not only classify images but also to detect several objects in garbage, based on the YOLOv8 framework, combined with Retrieval-Augmented Generation. This last framework is the greatest input of the thesis as it made the system more practical and realistic in the waste management solution. The detector could also detect and localize many waste items

and give confidence scores to each detected material rather than labeling an entire image with a single label. These identified labels were subsequently sent to the retrieval and generation module that generated recycling information as per curated knowledge on waste-management. Consequently, the end system was out of recognition and was transformed into a decision support system which can guide users in both identification and recycling action.

The achieved outcomes of the thesis prove the usefulness of the suggested scheme. The classification stages demonstrated a high predictive value whereas the final detection model demonstrated consistent multi-class recognition with significant values of precision, recall, F1-score, mAP, 50 and mAP, 50-95. Despite the fact that the object-detection framework is more demanding than the image-level classification, the metrics obtained showed the model acquired useful object-level representations and could be effectively used in mixed-waste setups. Moreover, the graph based assessment and confusion-matrix analysis, as well as interface level outputs proved that the system acted in a consistent, understandable and useful way. This demonstrates that the suggested framework is not only a valid one on an academic level, but also applicable in the practice of sustainable waste management.

The other significant result of the thesis is that it showed the usefulness of implementing a system that combines several AI paradigms. The work was not confined to the classification of images or object detection. Rather, it incorporated deep learning, hybrid feature extraction, localization of objects, retrieval based knowledge access and user-oriented output generation. This integration rendered the end system more holistic than traditional waste classification systems, as it featured the chain of waste identification to recycling directions. Thus, the thesis will not only bring an intelligent waste management support framework, but it will be more geared towards practical environmental requirements.

Finally, this thesis was able to show that artificial intelligence can be used significantly to enhance waste segregation and recycling support. The framework developed demonstrated that automated waste classification can enhance the need to use manual sorting, that hybrid architectures may be used to enhance the reliability of classification and that multi-object detection coupled with retrieval-based guidance can offer a more realistic and useful answer to sustainable waste management. Therefore, both technical

advancement and practical applicability were attained in the work, providing a solid base to the further research and advancement of the intelligent waste management systems.

5.2 Future Study :-

Despite the fact that the framework created brought about significant findings in waste classification, object detection, and recycling guidance, it can be concluded that there is a lot more to be fine-tuned and extended. A valuable research avenue in the future is to use bigger and more varied datasets. The quality and diversity of the training data has a strong impact on the performance of waste-recognition systems. Real-world waste-management scenarios can have objects shown in various lighting conditions, backgrounds, occlusions, and orientations. Thus, future research could be devoted to gathering more realistic waste images, making the data more diverse, and class balance, particularly in relation to those categories that are still hard to classify or identify. It would be beneficial to have a more data-rich dataset to enhance the reliability of classification and strength of object-detection in real-world conditions.

The other area of concern that needs to be studied further is the enhancement of the challenging classes of waste that demonstrated relatively low results in the detection phase. The result analysis showed that there were certain categories, paper, which were more challenging to the detector compared to the visually outstanding ones like glass and metal. This indicates that further research is required on data augmentation, class balancing, fine-grained annotation, and better feature-learning approaches of similar material as visual. Future studies can then focus on how to eliminate the confusion of classes and enhance the recall of those categories that are more difficult to distinguish with the background or other types of wastes.

Another direction to be extended is to use more sophisticated object-detection architectures and optimization methods. YOLOv8 resulted in stable and meaningful multi-object waste detection results, but new work can be done on better variants of detectors, more powerful backbone networks, or more lightweight to deploy to achieve higher localization accuracy and quicker inference. Model compression, pruning, as well as quantization might also be researched in such a way that the framework can be

utilized effectively on the edge devices, on a mobile platform, or on a waste-monitoring system embedded into a chip.

Further research can also be done on enhancing the Retrieval-Augmented Generation module. Within the present model, recycling instructions are created based on a curated knowledge base through retrieval and language generation. This can be furthered by building a bigger and more vibrant recycling knowledge base that encompasses the area-specific disposal regulations, local regulations, material handling directions, and sustainability policies. This would result in a more geographically and administratively suitable generated recycling advice that is more informative and useful. Additional enhancements to irrelevant outputs and guidance quality could also be due to improved retrieval strategies and the design of prompts.

The other potential research avenue in the future is the incorporation of real-time monitoring systems of waste. This can be expanded to live video streams to allow waste to be detected in real-life settings, as opposed to using only uploaded images. An extension that can be made is to place a camera in a garbage truck or waste collection unit to ensure that the system is able to monitor and track the kind of waste being dumped in real time. This would render the framework more appropriate to the application of municipal scale and would be capable of providing live waste analytics, tracking of recyclable materials and dynamic waste composition monitoring during transportation and collection.

Finally, future research may aim at larger and more detailed datasets, better balancing the classes, more robust detectors, augmented retrieval and generation methods, real-time video-based surveillance, smart-bin combination, multi-modal sensing, and better user interface. Such guidelines would enhance not only the technical functioning of the framework, but make it have a broader practical utility within the real-world sustainable waste management systems.

References:

- [1] D. Mewada et al., “Contamination Detection From Highly Cluttered Waste Scenes Using Computer Vision,” *IEEE Access*, vol. 12, pp.129434–129446, 2024, doi: 10.1109/ACCESS.2024.3456469.
- [2] A. A. Noman and N. J. Mila, “An Automated Effective Waste Classification and Sorting Using YOLOv8,” in *Proc. IEEE Int. Conf. Machine Learning and Applications (ICMLA)*, 2024, doi: 10.1109/ICMLA60862.2024.00112.
- [3] A. Author et al., “YOLOv8-based Waste Detection System for Recycling Plants: A Deep Learning Approach,” in *Proc. IEEE Int. Conf. Smart Systems and Advanced Computing (ICSSAS)*, 2023, doi: 10.1109/ICSSAS58688.2023.00052.
- [4] J. Wang et al., “Garbage Classification Recognition Based on Improved YOLOv8 Model,” in *Proc. IEEE Int. Conf. Image Processing (ICIP)*, 2024, doi: 10.1109/ICIP60790.2024.00118.
- [5] R. E. Mallouhy, H. Alqahtani, and S. Lardhi, “AI-Powered Standard Operating Procedure Generation and Optimization Using Large Language Models for Chemical Laboratory Applications,” *IEEE Access*, vol. 14, pp. 19383–19395, 2026, doi: 10.1109/ACCESS.2026.3660770.
- [6] R. A. Aral, M. O. Keskin, M. Kaya, and S. Hacıomeroglu, “Classification of TrashNet Dataset Based on Deep Learning Models,” in *Proc. IEEE Int. Conf. Big Data*, 2018, pp. 2058–2062, doi: 10.1109/Big-Data.2018.8622212.
- [7] N. Ramsurrun, G. Suddul, S. Armoogum, and R. Foogooa, “Recyclable Waste Classification Using Computer Vision and Deep Learning,” in *Proc. Zooming Innovation in Consumer Technologies Conf. (ZINC)*, 2021, doi: 10.1109/ZINC52049.2021.9499291.
- [8] F. S. Alsubaei, F. N. Al-Wesabi, and A. M. Hilal, “Deep Learning-Based Small Object Detection and Classification Model for Garbage Waste Management in Smart Cities and IoT Environment,” *Applied Sciences*, vol. 12, no. 5, p. 2281, 2022, doi: 10.3390/app12052281.

- [9] M. Malik et al., “Waste Classification for Sustainable Development Using Image Recognition with Deep Learning Neural Network Models,” *Sustainability*, vol. 14, no. 12, p. 7222, 2022, doi: 10.3390/su14127222.
- [10] N. Nnamoko and O. Barrowclough, “Solid Waste Image Classification Using Deep Convolutional Neural Networks,” *Infrastructures*, vol. 7, no. 4, p. 47, 2022.
- [11] Z. Yuan and J. Liu, “A Hybrid Deep Learning Model for Trash Classification Based on Deep Transfer Learning,” *Journal of Electrical and Computer Engineering*, vol. 2022, Article ID 7608794, 2022, doi: 10.1155/2022/7608794.
- [12] M. I. B. Ahmed et al., “Deep Learning Approach to Recyclable Products Classification: Towards Sustainable Waste Management,” *Sustainability*, vol. 15, no. 14, p. 11138, 2023, doi: 10.3390/su151411138.
- [13] N. Li and Y. Chen, “Municipal Solid Waste Classification and Real-Time Detection Using Deep Learning Methods,” *Urban Climate*, vol. 49, p. 101462, 2023, doi: 10.1016/j.uclim.2023.101462.
- [14] S. Poudel and P. Poudyal, “Classification of Waste Materials Using CNN Based on Transfer Learning,” in *Proc. Forum for Information Retrieval Evaluation (FIRE)*, 2023, doi: 10.1145/3574318.3574345.
- [15] A. Rahmatulloh et al., “WasteInNet: Deep Learning Model for Real- Time Identification of Various Types of Waste,” *Cleaner Waste Systems*, vol. 10, p. 100198, 2025, doi: 10.1016/j.clwas.2024.100198.
- [16] Hossen, M. M., Majid, M. E., Kashem, S. B. A., Khandakar, A., Nashbat, M., Ashraf, A., ... & Chowdhury, M. E. (2024). A reliable and robust deep learning model for effective recyclable waste classification. *IEEE Access*, 12, 13809-13821.
- [17] Rastari, M. A. M., Roslan, R., Hamzah, R., Teo, N. H. I., Shahbudin, F. E., & Samah, K. A. F. A. (2024, August). Recycle waste detection and classification model using YOLO-V8 for real-time waste management. In *2024 IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET)* (pp. 372-377). IEEE.

- [18] Sayem, F. R., Islam, M. S. B., Naznine, M., Nashbat, M., Hasan-Zia, M., Kunju, A. K. A., ... & Chowdhury, M. E. (2024). Enhancing waste sorting and recycling efficiency: robust deep learning-based approach for classification and detection. *Neural Computing and Applications*, 1-17.
- [19] Joseph, I. O., Wangare, N. L., Mahoque, T. P., Tewari, P., & Majumdar, S. (2023, July). E-Waste Intelligent Robotic Technology (EIRT): A Deep Learning approach for Electronic Waste Detection, Classification and Sorting. In 2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT) (pp. 1-6). IEEE.
- [20] Polchan, M., Pukao, A., Cheunban, T., & Sinthupuan, S. (2023, March). Waste Management System using Waste Classification on Mobile Application. In 2023 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECTI DAMT & NCON) (pp. 229-233). IEEE.
- [21] Thinakaran, R., Somasekar, J., Neerugatti, V., & Saran, P. G. (2024, August). Automated Waste Classification using Convolutional Neural Network. In 2024 14th International Conference on Software Technology and Engineering (ICSTE) (pp. 169-173). IEEE.
- [22] Jain, E., & Kumar, R. (2024, November). Harnessing Deep Learning for Sustainable Waste Classification: An Innovative Approach Using MobileNetV2 on a Diverse Bag Dataset. In 2024 International Conference on Cybernation and Computation (CYBERCOM) (pp. 335-339). IEEE.
- [23] Yang, M., Thung, G., & Tseng, M. (2022). Classification of trash for recyclability status using deep learning. *Resources, Conservation and Recycling*, 175, 105873.
- [24] Mittal, G., Yagnik, K. B., Garg, M., & Krishnan, N. C. (2021). SpotGarbage: Smartphone app to detect garbage using deep learning. *ACM International Joint Conference on Pervasive and Ubiquitous Computing*, 940–945.
- [25] Rad, M. S., von Kaenel, J., Droux, A., & Thiran, J. P. (2017). A computer vision system to localize and classify wastes on the streets. *IEEE Transactions on Intelligent Transportation Systems*, 18(12), 311

Appendix

The technical implementation of the garbage detection and recycling guidance framework that is based on YOLOv8 is provided in the given appendix. The appendix includes only the key stages of the notebook in order to include only the most relevant implementation details. The general pipeline starts with configuring the dataset and training the model, followed by the waste detection and visualizing the results, interactive testing, and lastly recycling guidance generation. The framework will identify a variety of waste types based on an uploaded image and give material-specific recycling suggestions based on this input using a retrieval and generation module.

Stage 1: Preparation and Configuration of Dataset.

High-level Logic: configuration of datasets and YAML.

The first step is to set the path to the garbage dataset and a dataset configuration file needed by YOLOv8. The dataset is structured in YOLO format, with individual folders being kept of training, validation and testing images and their annotation files. This setup is significant since YOLOv8 needs a structured YAML file filled with dataset paths, amount of classes and class names prior to the commencement of training.

The dataset employed within the framework comprises of six types of waste paper, plastic, glass, metal, cardboard and biodegradable waste. The data is split to 70% training, 20% validation and 10% testing which facilitates systematic learning, model monitoring and fair evaluation. All images are scaled to 640 x 640, so that they fit the default size of the detector.

```
dataset_path = "/kaggle/input/datasets/jitinvenkatasai/major-project"

if not os.path.exists(dataset_path):
    print(f"Path {dataset_path} not found. Searching for 'major-project'...")
    # Fallback to typical Kaggle path
    dataset_path = "/kaggle/input/major-project"

print(f"Using dataset path: {dataset_path}")
```

Figure A.1 : Dataset Configuration

This step is necessary to make sure that the detector is able to access the training data properly and learn the target waste types.

Stage 2: YOLOv8 Model First Training and Initialization.

Primary Logic: YOLOv8 loading and training a detector.

Once the datasets are prepared, the YOLOv8 model is pretrained with the pretrained weights. Transfer learning is applied in such a way that the model initializes with the already learned visual features and then trains them to the waste-detection task. This renders training more effective and assists the detector to converge quicker than training

by chance. The training is implemented in a GPU-enabled Kaggle Notebook setup, which can be used to experiment with deep learning and develop models that are reproducible.

The detector is implemented with a fixed number of epochs using an input size of 640 and optimized by a batch. In training, YOLOv8 is trained to detect waste objects with bounding boxes as well as classify it into the supported waste types. The training phase is the heart of visual-recognition aspect of the entire system.

```
model = YOLO('yolov8n.pt')

results = model.train(
    data=working_yaml_path,
    epochs=30,
    imgsz=640,
    batch=16,
    project="Garbage_Detection",
    name="run_1",
    seed=42
)
print("Training complete! Weights are saved in /kaggle/working/Garbage_Detection/run_1/weights/")
```

Figure A.2 : YOLOv8 Training

This step allows the system to transition out of basic image recognition on the way to useful multi-object waste detection

Stage 3: Visualization and Detection Analysis of performance.

Main Logic: generation and checking of the graph.

After training, the notebook produces the significant performance outputs needed in analyzing results. These are mAP curve, training box loss, validation box loss, F1-confidence curve and confusion matrix. These graphs are significant since they give an understanding of how the detector was improved in the course of time, and the extent to which it generalized to unknown data.

The overall results reported for the detector show Precision = 0.619, Recall = 0.451, F1-score = 0.522, mAP@50 = 0.535, and mAP@50-95 = 0.372 The mAP curve rose gradually throughout the training, and the box losses dropped, which means that the quality of localization improved and there was no instability in learning behaviour.

```

import os
from IPython.display import Image, display

run_dir = "/kaggle/working/runs/detect/Garbage_Detection/run_1"

def show_graph(filename):
    filepath = os.path.join(run_dir, filename)
    if os.path.exists(filepath):
        display(Image(filename=filepath, width=800))
    else:
        print(f"Could not find {filename}")

```

Figure A.3 : Result Visualization

This step facilitates the interpretation of the performance of the detector and makes the results that will be used in the results chapter of the thesis.

Stage 4: Interactive Waste Image Detection

Primary Logic: inference on uploaded images based on trained weights which are best.

The optimal weights of the models are loaded after training and testing and then tested practically. The notebook has a feature of an interactive mechanism whereby a user can post an image and can then do a direct run of waste detection. After uploading the image, the trained YOLOv8 model makes an inference and provides the detected waste objects and their bounding boxes, class labels, and confidence scores.

This phase is important as it shows the real-life performance of the detector. Rather than staying just a training notebook, the implementation will be a working test environment where the detection model can process new images of waste as they are given by the user. This validates the usefulness of the framework in practice beyond offline testing.

```

best_weights = "/kaggle/working/runs/detect/Garbage_Detection/run_1/weights/best.pt"
model = YOLO(best_weights)

uploader = widgets.FileUpload(
    accept='image/*',
    multiple=False,
    description='Upload Image'
)

```

Figure A.4 : Best Weight Loading and Upload Interface

Stage 5: Visualization of Detection and Bounding Box.

Main Logic: multi-object output generation.

After processing the image that is uploaded, the detector produces object-level outputs. Such outputs are the position of the waste objects, the estimated class names and confidence scores of each prediction. Bounding boxes are used to highlight the identified objects to enable the results to be visually understandable to the user.

It is one of the key phases of the implementation as it demonstrates the key practical power of YOLOv8. The detector is not only able to detect several waste objects on a single image, but also localize each of them individually, unlike a simple image classifier. This renders the framework more applicable to real waste-management settings where waste products tend to be found mixed in the same scenes.

```
for box in results[0].boxes:  
    cls_id = int(box.cls[0])  
    conf = float(box.conf[0])
```

Figure A.5 : Detection Result Structure

Stage 6: Recycling Advice according to RAG and LLM.

Primary Logic: forwarding of detected-labels to retrieval and generation pipe.

Once the YOLOv8 model has made predictions on the waste labels, the identified classes are sent to the recycling guidance module. The identified categories of waste are then considered query inputs in the retrieval stage where the system traverses a curated knowledge base of category-specific recycling guidelines. The content retrieved is then sent to the language generation phase where it is translated to a structured and easy-to-use recycling summary.

The output that will be generated can comprise of cleaning instructions, sorting, separation, recycling procedure and correct disposal of the identified materials. This step goes further to expand the framework to include object detection and convert it into a viable waste-management support system. That is, the system does not cease with the identification of the waste item, but informs the user what he or she should do with the waste once it has been detected.

The given step is especially significant as it integrates the computer vision with knowledge-based support and turns the end framework more educative and user-friendly than a traditional detection-only system.

Stage 7: Integrated User-Level Output

Primary Logic: waste management support in an end-to-end approach.

The end product of the notebook-based structure is a combined waste-management product which comprises:

- the uploaded image,
- the identified waste has bounding boxes,
- the waste assigns the values of confidence, and
- the recycling instructions created.

This last level indicates the entire working behavior of the framework as seen by the user. The uploaded image is initially processed by the visual analysis of YOLOv8, and the labels identified and processed by the retrieval and generation module, and the final output is presented in a readable and comprehensible format. This not only qualifies the framework as a detection system, but also a viable support mechanism towards sustainable waste management.

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